

Electromagnetic Scattering

Jake W. Liu

September 13, 2026

- 1 Cross Sections
- 2 Cylindrical Waves
 - Cylindrical Wave Solution
 - Cylindrical Wave Transformation
- 3 Scattering from PEC Cylinders
 - TM (E-Polarized) Incidence
 - TE (H-Polarized) Incidence
- 4 Spherical Waves
 - Spherical Wave Solution
 - Spherical Wave Transformation
- 5 Scattering from Dielectric Spheres

Chapter Overview

This chapter provides an overview of electromagnetic scattering, a fundamental phenomenon that occurs when a wave encounters an obstacle. We will begin by defining the concept of scattering cross-section, a key parameter used to quantify the strength and angular distribution of scattered fields. The chapter will then delve into the analytical solutions for scattering by canonical objects, starting with two-dimensional problems involving PEC cylinders, and then extending the analysis to three-dimensional scattering by dielectric spheres, commonly known as Mie theory.

Table of contents

1 Cross Sections

2 Cylindrical Waves

- Cylindrical Wave Solution
- Cylindrical Wave Transformation

3 Scattering from PEC Cylinders

- TM (E-Polarized) Incidence
- TE (H-Polarized) Incidence

4 Spherical Waves

- Spherical Wave Solution
- Spherical Wave Transformation

5 Scattering from Dielectric Spheres

Cross Sections

In the analysis of scattering problems, the concept of cross section is frequently employed to quantitatively characterize how an object interacts with and scatters electromagnetic waves in the far field. In three dimensions, a cross section measures an effective area that intercepts and reradiates incident energy; in a two-dimensional problem invariant along one axis, the corresponding quantity is an effective transverse width, with power understood per unit invariant length. In this chapter, various types of cross sections are defined. These quantities prove particularly useful when comparing the scattering behavior of different objects or materials.

Unless stated otherwise, the exterior medium in this chapter is homogeneous, isotropic, and lossless, with real positive ϵ and μ , wavenumber $k = \omega\sqrt{\mu\epsilon} > 0$, and wave impedance $\eta = \sqrt{\mu/\epsilon} > 0$. This assumption is needed below when the real, time-averaged radial power density is written as $|E|^2/(2\eta)$. A lossy host requires $\text{Re}(\vec{E} \times \vec{H}^*)/2$ and a correspondingly modified normalization.

A scattering problem typically involves determining the field scattered by an obstacle. The incident field is often represented as a plane wave propagating in the $\hat{i}$ direction:

$$\vec{E}_i = \vec{E}_0 e^{-ik\hat{i}\cdot\vec{r}}, \quad \hat{i} \cdot \vec{E}_0 = 0. \quad (4.1.1)$$

Cross Sections (cont.)

Physically, this plane-wave representation approximates the far field produced by a radar antenna located at a large distance from the target. In this configuration, the wavefronts arriving at the target are effectively planar, and the observation (receiving) antenna is assumed to be in the target's far-field zone. A localized three-dimensional target then produces a radiating spherical wave; the ideal two-dimensional, translation-invariant model used for an infinite cylinder instead produces a cylindrical wave. The scattered field is defined in either case by

$$\vec{E}_s = \vec{E}_t - \vec{E}_i. \quad (4.1.2)$$

The scattered field is then fixed by the matching conditions imposed on the total electric or magnetic field at the surface of the scatterer.

In the far field, let $E_0 = |\vec{E}_0|$, and let $\vec{f}(\hat{s}, \hat{i})$ denote the vector scattering-amplitude function in observation direction $\hat{s}$ for incidence direction $\hat{i}$. With these quantities defined, the scattered field takes the form

$$\vec{E}_s = \begin{cases} E_0 \frac{e^{-ikr}}{r} \vec{f}(\hat{s}, \hat{i}) & (3D), \\ E_0 \frac{e^{-ik\rho}}{\sqrt{\rho}} \vec{f}(\hat{s}, \hat{i}) & (2D). \end{cases} \quad (4.1.3)$$

Cross Sections (cont.)

From (1.5.7), the incident and scattered power flux densities are evaluated by noting that each field is locally a plane wave, so its magnetic field follows from Faraday's law $\nabla \times \vec{E} = -i\omega\mu\vec{H}$. For the incident wave (4.1.1), $\nabla \times \vec{E}_i = -ik\hat{i} \times \vec{E}_i$, and since $k/(\omega\mu) = 1/\eta$,

$$\vec{H}_i = \frac{1}{\eta} \hat{i} \times \vec{E}_i.$$

Substituting this into the Poynting vector and applying the identity $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$,

$$\begin{aligned} \vec{S}_i &= \frac{1}{2} \vec{E}_i \times \vec{H}_i^* = \frac{1}{2\eta} \vec{E}_i \times (\hat{i} \times \vec{E}_i^*) \\ &= \frac{1}{2\eta} \left[\hat{i} (\vec{E}_i \cdot \vec{E}_i^*) - \vec{E}_i^* (\vec{E}_i \cdot \hat{i}) \right]. \end{aligned}$$

The plane wave is transverse, $\hat{i} \cdot \vec{E}_i = 0$, so the second term vanishes; writing $|\vec{E}_i|^2 \equiv \vec{E}_i \cdot \vec{E}_i^*$, only the longitudinal part remains. The scattered field is locally plane in

Cross Sections (cont.)

the far zone, whether its radial spreading is spherical in 3D or cylindrical in 2D, so the same steps apply with $\hat{i} \rightarrow \hat{s}$ and $\vec{E}_i \rightarrow \vec{E}_s$. This gives

$$\vec{S}_i = \frac{1}{2}(\vec{E}_i \times \vec{H}_i^*) = \frac{|E_i|^2}{2\eta} \hat{i}, \quad \vec{S}_s = \frac{1}{2}(\vec{E}_s \times \vec{H}_s^*) = \frac{|E_s|^2}{2\eta} \hat{s}. \quad (4.1.4)$$

The differential scattering cross section is defined as:

$$\sigma_d(\hat{s}, \hat{i}) = \begin{cases} \lim_{r \rightarrow \infty} r^2 \frac{|\vec{S}_s|}{|\vec{S}_i|} & (3D) \\ \lim_{\rho \rightarrow \infty} \rho \frac{|\vec{S}_s|}{|\vec{S}_i|} & (2D) \end{cases} = |\vec{f}(\hat{s}, \hat{i})|^2. \quad (4.1.5)$$

To evaluate this ratio, note that the incident plane wave (4.1.1) has $|\vec{E}_i| = E_0$, so that (1.5.7) gives

$$|\vec{S}_i| = \frac{|\vec{E}_i|^2}{2\eta} = \frac{E_0^2}{2\eta}.$$

Cross Sections (cont.)

For the scattered field, $\hat{s}$ is a unit vector and the lossless radial phase factor has unit modulus ($|e^{-ikr}| = |e^{-ik\rho}| = 1$), so the far-field form yields

$$|\vec{E}_s|^2 = \frac{E_0^2}{r^2} |\vec{f}|^2 \quad (3D), \quad |\vec{E}_s|^2 = \frac{E_0^2}{\rho} |\vec{f}|^2 \quad (2D),$$

where $|\vec{f}|^2 = \vec{f} \cdot \vec{f}^*$. Hence

$$|\vec{S}_s| = \frac{|\vec{E}_s|^2}{2\eta} = \begin{cases} \frac{E_0^2 |\vec{f}|^2}{2\eta r^2} & (3D), \\ \frac{E_0^2 |\vec{f}|^2}{2\eta \rho} & (2D). \end{cases}$$

Forming the ratio, the $1/(2\eta)$ factors and the radial decay cancel,

$$r^2 \frac{|\vec{S}_s|}{|\vec{S}_i|} = |\vec{f}|^2 \quad (3D), \quad \rho \frac{|\vec{S}_s|}{|\vec{S}_i|} = |\vec{f}|^2 \quad (2D),$$

which is already independent of r (or ρ), so the limit is immediate, establishing the last equality, $\sigma_d = |\vec{f}|^2$, in (4.1.5). The dimensions follow directly from the far-field

Cross Sections (cont.)

normalization in (4.1.3): $\vec{f}$ has dimensions of length in 3D and of the square root of length in 2D. Consequently, σ_d has dimensions of area in 3D but of length in 2D. The bistatic radar cross section (RCS) is then defined as:

$$\sigma_{bi}(\hat{s}, \hat{i}) = \begin{cases} 4\pi\sigma_d(\hat{s}, \hat{i}) & (3D), \\ 2\pi\sigma_d(\hat{s}, \hat{i}) & (2D). \end{cases} \quad (4.1.6)$$

In two dimensions, σ_{bi} is also referred to as the echo width.

In 3D, the bistatic RCS represents a hypothetical area that, if illuminated by the incident power density and reradiating that power isotropically, would produce the same received power density as the actual target. In 2D, the analogous echo width is a hypothetical transverse width, not an area; incident and scattered powers are understood per unit invariant cylinder length. Both quantities are defined to be independent of the distance between the radar and the target, but they strongly depend on the incidence angle, observation angle, polarization, frequency, and the target's material and shape. Specifically, the monostatic or backscattering RCS refers to the scenario where the radar transmitter and receiver are co-located, thereby measuring the power reflected directly back toward the source:

$$\sigma_{mono}(\hat{i}) = \sigma_{bi}(-\hat{i}, \hat{i}). \quad (4.1.7)$$

Cross Sections (cont.)

Finally, the total scattering cross section, which quantifies the amount of incident power scattered by an object in all directions, is defined as:

$$\sigma_{sca} = \begin{cases} \int_{4\pi} \sigma_d d\Omega = \int_{4\pi} r^2 \frac{|\vec{S}_s|}{|\vec{S}_i|} d\Omega & (3D), \\ \int_{2\pi} \sigma_d d\phi = \int_{2\pi} \rho \frac{|\vec{S}_s|}{|\vec{S}_i|} d\phi & (2D), \end{cases} \quad (4.1.8)$$

where $d\Omega$ represents the differential solid angle.

Table of contents

- 1 Cross Sections
- 2 Cylindrical Waves
 - Cylindrical Wave Solution
 - Cylindrical Wave Transformation
- 3 Scattering from PEC Cylinders
 - TM (E-Polarized) Incidence
 - TE (H-Polarized) Incidence
- 4 Spherical Waves
 - Spherical Wave Solution
 - Spherical Wave Transformation
- 5 Scattering from Dielectric Spheres

Cylindrical Waves

This section focuses on the wave and Helmholtz equations in cylindrical coordinates and on the corresponding cylinder-scattering problem; spherical waves and sphere scattering are treated separately in Section 4.4. As noted in Section 1.4, in a source-free region, the scalar Helmholtz equation (1.4.16) applies to all three Cartesian components. Of the physical components resolved in the cylindrical basis, only the z component has a position-independent unit vector and therefore satisfies that scalar equation directly, without curvature-induced coupling to another cylindrical component. The Cartesian x and y components still satisfy scalar Helmholtz equations; they are simply not the physical radial and azimuthal components. In spherical coordinates, none of the local unit vectors are constant, necessitating the additional analysis in Section 4.4.

Cylindrical Wave Solution

As previously mentioned, the vector Helmholtz equation can be simplified to a scalar equation for the z-component in cylindrical coordinates:

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{\partial^2 \psi}{\partial z^2} + k^2 \psi = 0. \quad (4.2.1)$$

This equation can be solved using the method of separation of variables, by assuming a solution of the form:

$$\psi = B(\rho)\Phi(\phi)Z(z). \quad (4.2.2)$$

Substituting this into (4.2.1) yields:

$$\frac{1}{B\rho} \frac{d}{d\rho} \left(\rho \frac{dB}{d\rho} \right) + \frac{1}{\Phi\rho^2} \frac{d^2\Phi}{d\phi^2} + \frac{1}{Z} \frac{d^2Z}{dz^2} + k^2 = 0. \quad (4.2.3)$$

Since the variable z appears only in the third term of (4.2.3), we can separate it, leading to an equation similar to the rectangular case:

$$\frac{d^2Z}{dz^2} + k_z^2 Z = 0. \quad (4.2.4)$$

Cylindrical Wave Solution (cont.)

For $k_z \neq 0$, the elementary solutions for Z are:

$$Z = e^{\pm ik_z z}. \quad (4.2.5)$$

At $k_z = 0$ these two exponentials coalesce; the full solution of $Z'' = 0$ is $Z = A + Bz$. The two-dimensional cylinder-scattering problem below is invariant along z and selects the constant solution. Now, let

$$k_\rho^2 = k^2 - k_z^2. \quad (4.2.6)$$

Multiplying (4.2.3) by ρ^2 , we obtain:

$$\frac{\rho}{B} \frac{d}{d\rho} \left(\rho \frac{dB}{d\rho} \right) + \frac{1}{\Phi} \frac{d^2\Phi}{d\phi^2} + k_\rho^2 \rho^2 = 0. \quad (4.2.7)$$

Similarly, the middle term of (4.2.7), $\frac{1}{\Phi} \frac{d^2\Phi}{d\phi^2}$, depends only on ϕ , whereas the remaining terms depend only on ρ . Since their sum must vanish for all ρ and ϕ independently, each part must equal a constant. We write the azimuthal part as a negative constant,

$$\frac{1}{\Phi} \frac{d^2\Phi}{d\phi^2} = -\nu^2,$$

Cylindrical Wave Solution (cont.)

so that the radial remainder equals $+\nu^2$, that is,

$$\frac{\rho}{B} \frac{d}{d\rho} \left(\rho \frac{dB}{d\rho} \right) + k_\rho^2 \rho^2 = \nu^2.$$

The negative sign is required for Φ to be single-valued over 2π : it gives the oscillatory solution $\Phi = e^{\pm i\nu\phi}$, whereas a positive constant would yield real exponentials that cannot be periodic. The azimuthal part thus reproduces the equation for Φ in (4.2.8),

$$\frac{d^2\Phi}{d\phi^2} + \nu^2\Phi = 0, \quad (4.2.8)$$

which has the elementary solution of:

$$\Phi = e^{\pm i\nu\phi}. \quad (4.2.9)$$

By the same periodicity, $\nu = n$ is an integer constant. Consequently, (4.2.7) simplifies to:

$$\frac{d^2B}{d\rho^2} + \frac{1}{\rho} \frac{dB}{d\rho} + \left(k_\rho^2 - \frac{\nu^2}{\rho^2} \right) B = 0. \quad (4.2.10)$$

Cylindrical Wave Solution (cont.)

This is known as the Bessel equation of order ν . For $k_\rho \neq 0$, its solutions are the Bessel functions described below. The cutoff case $k_\rho = 0$ must be taken separately: the radial equation is then an Euler equation with

$$\begin{aligned} B(\rho) &= C\rho^\nu + D\rho^{-\nu}, & \nu > 0, \\ B(\rho) &= C + D \ln \rho, & \nu = 0. \end{aligned}$$

Thus one must not obtain a cutoff basis by directly substituting $k_\rho = 0$ into $J_\nu(k_\rho\rho)$ and $Y_\nu(k_\rho\rho)$. The propagating wave and cylinder-scattering formulas below use real $k_\rho > 0$. For $\nu \geq 0$, the solution to (4.2.10) that remains finite at $\rho = 0$ is known as the Bessel function of the first kind:

$$J_\nu(k_\rho\rho) = \sum_{m=0}^{\infty} \frac{(-1)^m (k_\rho\rho/2)^{\nu+2m}}{m! \Gamma(m + \nu + 1)}. \quad (4.2.11)$$

When ν is a nonnegative integer, $\Gamma(m + \nu + 1) = (m + \nu)!$, recovering the factorial form. Because the differential equation depends on ν^2 , we may take $\nu \geq 0$. For

Cylindrical Wave Solution (cont.)

noninteger $\nu > 0$, a second independent solution, $J_{-\nu}(k_\rho \rho)$, exists but diverges at $\rho = 0$. If ν is an integer, denoted by n , then:

$$J_{-n}(k_\rho \rho) = (-1)^n J_n(k_\rho \rho). \quad (4.2.12)$$

A second independent solution, called the Neumann function, is constructed as:

$$Y_n(k_\rho \rho) = \lim_{\nu \rightarrow n} \frac{J_\nu(k_\rho \rho) \cos(\nu\pi) - J_{-\nu}(k_\rho \rho)}{\sin(\nu\pi)}. \quad (4.2.13)$$

The Hankel functions of the first and second kind are combinations of the Bessel and Neumann functions:

$$H_n^{(1)}(k_\rho \rho) = J_n(k_\rho \rho) + iY_n(k_\rho \rho), \quad (4.2.14)$$

$$H_n^{(2)}(k_\rho \rho) = J_n(k_\rho \rho) - iY_n(k_\rho \rho). \quad (4.2.15)$$

For large arguments ($k_\rho \rho \gg 1$), the Hankel functions can be asymptotically approximated as:

$$H_n^{(1)}(k_\rho \rho) \approx \sqrt{\frac{2}{\pi k_\rho \rho}} e^{i(k_\rho \rho - n\pi/2 - \pi/4)}, \quad (4.2.16)$$

Cylindrical Wave Solution (cont.)

$$H_n^{(2)}(k_\rho \rho) \approx \sqrt{\frac{2}{\pi k_\rho \rho}} e^{-i(k_\rho \rho - n\pi/2 - \pi/4)}. \quad (4.2.17)$$

For real $k_\rho > 0$, these asymptotics, with the $e^{i\omega t}$ time convention, represent incoming and outgoing traveling cylindrical waves, respectively. If k_ρ is imaginary or genuinely complex, the radial behavior is evanescent or attenuated and this traveling-wave phase interpretation must be replaced by the corresponding analytic continuation. A few remarks justify the construction above. The relation $J_{-n} = (-1)^n J_n$ shows that, for integer order, J_n and J_{-n} are not independent: their Wronskian

$$W_x\{J_\nu, J_{-\nu}\}(x) = -\frac{2 \sin(\nu\pi)}{\pi x} \quad \text{with } x = k_\rho \rho,$$

where $W_x\{f, g\} = f(dg/dx) - (df/dx)g$, vanishes as $\nu \rightarrow n$, so a single power series cannot span the two-dimensional solution space of the second-order equation (4.2.10). The Neumann function (4.2.13) restores the missing second solution. At integer order its defining quotient is indeterminate, since

$$J_n \cos(n\pi) - J_{-n} = (-1)^n J_n - (-1)^n J_n = 0, \quad \sin(n\pi) = 0,$$

Cylindrical Wave Solution (cont.)

a 0/0 form; the limit is evaluated by L'Hôpital's rule, differentiating numerator and denominator with respect to ν . The large-argument forms (4.2.16) and (4.2.17) are standard asymptotic results [9, 1]. To see this, restore the suppressed time factor $e^{i\omega t}$ and retain only the ρ -dependent part of the phase (the constant $-n\pi/2 - \pi/4$ does not affect the direction of propagation). Their combined space-time phases are

$$H_n^{(1)}(k_\rho \rho) e^{i\omega t} \propto \frac{e^{i(\omega t + k_\rho \rho)}}{\sqrt{\rho}},$$
$$H_n^{(2)}(k_\rho \rho) e^{i\omega t} \propto \frac{e^{i(\omega t - k_\rho \rho)}}{\sqrt{\rho}}.$$

A surface of constant phase satisfies $\omega t \pm k_\rho \rho = \text{const.}$ For $H_n^{(1)}$, the $+$ sign forces ρ to *decrease* as t increases, so the wave converges toward the axis (incoming); for $H_n^{(2)}$, the $-$ sign forces ρ to *increase* with t , so the wave diverges outward (outgoing). This is why $H_n^{(2)}$ is the appropriate choice for a scattered field radiating away from the cylinder.

Cylindrical Wave Transformation

Consider a z-polarized plane wave propagating in the x-direction:

$$\vec{E} = \hat{z} E_0 e^{-ikx} = \hat{z} E_0 e^{-ik\rho \cos \phi}. \quad (4.2.18)$$

We can expand this exponential term into an infinite sum of cylindrical waves:

$$e^{-ik\rho \cos \phi} = \sum_{n=-\infty}^{\infty} p_n^{(\text{cyl})} J_n(k\rho) e^{in\phi}. \quad (4.2.19)$$

To determine the temporary plane-wave coefficients $p_n^{(\text{cyl})}$, first derive the nontrivial integral rather than assume it. The Bessel generating function is

$$\exp\left[\frac{z}{2}(t - t^{-1})\right] = \sum_{n=-\infty}^{\infty} J_n(z) t^n.$$

Setting $t = -ie^{i\phi}$ gives $t - t^{-1} = -2i \cos \phi$ and therefore

$$e^{-iz \cos \phi} = \sum_{n=-\infty}^{\infty} i^{-n} J_n(z) e^{in\phi}.$$

Cylindrical Wave Transformation (cont.)

Multiplying this generating-function identity by $e^{-im\phi}$ and using Fourier orthogonality gives the first of the two identities used below; the second is the orthogonality relation itself:

$$\int_0^{2\pi} e^{-i(k\rho \cos \phi + m\phi)} d\phi = 2\pi i^{-m} J_m(k\rho), \quad (4.2.20)$$

$$\int_0^{2\pi} e^{i(m-n)\phi} d\phi = 2\pi \delta_{mn}. \quad (4.2.21)$$

Multiplying (4.2.19) by $e^{-im\phi}$ and integrating over ϕ from 0 to 2π projects out a single term: the left-hand side is evaluated by the first identity, while the second identity collapses the right-hand sum to its $n = m$ term,

$$\begin{aligned} \int_0^{2\pi} e^{-ik\rho \cos \phi} e^{-im\phi} d\phi &= \sum_{n=-\infty}^{\infty} p_n^{(\text{cyl})} J_n(k\rho) \\ &\quad \times \int_0^{2\pi} e^{i(n-m)\phi} d\phi, \end{aligned}$$

Cylindrical Wave Transformation (cont.)

that is, $2\pi i^{-m} J_m(k\rho) = 2\pi p_m^{(\text{cyl})} J_m(k\rho)$ for every ρ . The function $J_m(k\rho)$ is not identically zero, so choose any ρ outside its isolated zero set. Cancellation there gives a coefficient identity that is independent of ρ and therefore applies to the entire expansion:

$$p_m^{(\text{cyl})} = i^{-m}. \quad (4.2.22)$$

Thus, we obtain the following expansion:

$$E_z = E_0 e^{-ikx} = E_0 e^{-ik\rho \cos \phi} = E_0 \sum_{n=-\infty}^{\infty} i^{-n} J_n(k\rho) e^{in\phi}. \quad (4.2.23)$$

This process is known as the cylindrical wave transformation, which expands a plane wave into a sum of cylindrical waves.

Table of contents

- 1 Cross Sections
- 2 Cylindrical Waves
 - Cylindrical Wave Solution
 - Cylindrical Wave Transformation
- 3 Scattering from PEC Cylinders
 - TM (E-Polarized) Incidence
 - TE (H-Polarized) Incidence
- 4 Spherical Waves
 - Spherical Wave Solution
 - Spherical Wave Transformation
- 5 Scattering from Dielectric Spheres

Scattering from PEC Cylinders

This section examines the scattering of a plane wave by a PEC cylinder of radius a , aligned along the z -axis. Two distinct incident polarizations are considered with respect to the cylinder's axis: E-polarization (transverse magnetic to z , or TM) and H-polarization (transverse electric to z , or TE). We derive both cases below.

TM (E-Polarized) Incidence

Consider a plane wave propagating in the x -direction. Using the expansion from (4.2.23), the incident field can be expressed as:

$$\vec{E}_i = E_0 e^{-ikx} \hat{z} = \hat{z} E_0 \sum_{n=-\infty}^{\infty} i^{-n} J_n(k\rho) e^{in\phi}. \quad (4.3.1)$$

We assume the scattered field takes the form:

$$\vec{E}_s = \hat{z} E_0 \sum_{n=-\infty}^{\infty} i^{-n} a_n H_n^{(2)}(k\rho) e^{in\phi}. \quad (4.3.2)$$

The Hankel function of the second kind, $H_n^{(2)}(k\rho)$, is used here to represent outgoing propagating waves. Applying the matching condition that the total tangential electric field E_z^t must be zero at the cylinder surface ($\rho = a$), we have:

$$E_z^t = E_0 \sum_{n=-\infty}^{\infty} i^{-n} [J_n(ka) + a_n H_n^{(2)}(ka)] e^{in\phi} = 0. \quad (4.3.3)$$

TM (E-Polarized) Incidence (cont.)

Because the functions $\{e^{in\phi}\}$ are mutually orthogonal on $[0, 2\pi]$, the only way for the sum in (4.3.3) to vanish for every ϕ is for each term to vanish individually. To extract the m -th term, we multiply (4.3.3) by $e^{-im\phi}$ and integrate over ϕ from 0 to 2π , using $\int_0^{2\pi} e^{i(m-n)\phi} d\phi = 2\pi\delta_{mn}$:

$$E_0 \sum_{n=-\infty}^{\infty} i^{-n} [J_n(ka) + a_n H_n^{(2)}(ka)] 2\pi\delta_{mn} = 0,$$

which collapses the sum to the single surviving term $n = m$,

$$2\pi E_0 i^{-m} [J_m(ka) + a_m H_m^{(2)}(ka)] = 0.$$

Since $E_0 \neq 0$ and $i^{-m} \neq 0$, the bracket itself must vanish for every m ,

$$J_m(ka) + a_m H_m^{(2)}(ka) = 0.$$

Relabeling $m \rightarrow n$ and solving for the coefficient then gives

$$a_n = -\frac{J_n(ka)}{H_n^{(2)}(ka)}. \quad (4.3.4)$$

TM (E-Polarized) Incidence (cont.)

Substituting this back, the scattered field is:

$$\vec{E}_s = -\hat{z}E_0 \sum_{n=-\infty}^{\infty} i^{-n} \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(k\rho) e^{in\phi}. \quad (4.3.5)$$

And the total field is:

$$\vec{E}_t = \hat{z}E_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n(k\rho) - \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(k\rho) \right] e^{in\phi}. \quad (4.3.6)$$

We first isolate the ρ - and n -dependence of the asymptotic Hankel function by writing (4.2.17) as

$$H_n^{(2)}(k\rho) \approx \sqrt{\frac{2}{\pi k\rho}} e^{-ik\rho} e^{in\pi/2} e^{i\pi/4},$$

where we used $e^{-i(k\rho - n\pi/2 - \pi/4)} = e^{-ik\rho} e^{in\pi/2} e^{i\pi/4}$. Substituting this into the scattered field (4.3.5) and pulling the n -independent factors out of the sum yields

$$\vec{E}_s \approx -\hat{z}E_0 \sqrt{\frac{2}{\pi k\rho}} e^{i\pi/4} e^{-ik\rho} \sum_{n=-\infty}^{\infty} i^{-n} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\pi/2} e^{in\phi}.$$

TM (E-Polarized) Incidence (cont.)

The n -dependent phase from the asymptotic form is $e^{in\pi/2} = i^n$, which cancels the i^{-n} already present in each term,

$$i^{-n} e^{in\pi/2} = i^{-n} i^n = 1,$$

so the summand reduces to $[J_n(ka)/H_n^{(2)}(ka)] e^{in\phi}$. Finally, the constant phase is absorbed into the radical using $e^{i\pi/4} = \sqrt{i}$, giving $\sqrt{2/(\pi k\rho)} \sqrt{i} = \sqrt{2i/(\pi k\rho)}$. Collecting the outgoing $e^{-ik\rho}/\sqrt{\rho}$ dependence, we obtain

$$\vec{E}_s \approx -\hat{z} E_0 \sqrt{\frac{2i}{\pi k}} \frac{e^{-ik\rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi},$$

which is precisely the 2D far-field form $E_0 (e^{-ik\rho}/\sqrt{\rho}) \vec{f}(\hat{s}, \hat{i})$ of (4.1.3), with the scattering amplitude

$$\vec{f}(\hat{s}, \hat{i}) = -\hat{z} \sqrt{\frac{2i}{\pi k}} \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi}.$$

TM (E-Polarized) Incidence (cont.)

Writing $\sqrt{i} = e^{i\pi/4}$ gives the equivalent phase-factored form $\vec{f} = -\hat{z} \sqrt{2/(\pi k)} e^{i\pi/4} \sum_{n=-\infty}^{\infty} [J_n(ka)/H_n^{(2)}(ka)] e^{in\phi}$. Since this unit-magnitude factor drops out of the modulus, the differential cross section $\sigma_d = |\vec{f}|^2$ from (4.1.5) becomes

$$\sigma_d = |\vec{f}|^2 = \frac{2}{\pi k} \left| \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi} \right|^2.$$

Inserting this into the two-dimensional case of (4.1.6), $\sigma_{bi} = 2\pi \sigma_d$, the factor of π cancels and the prefactor reduces to $4/k$, yielding

$$\sigma_{bi} = \frac{4}{k} \left| \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi} \right|^2. \quad (4.3.7)$$

This result is often normalized by the free-space wavelength $\lambda = 2\pi/k$, which reduces the prefactor $4/k$ to

$$\frac{1}{\lambda} \cdot \frac{4}{k} = \frac{k}{2\pi} \cdot \frac{4}{k} = \frac{4}{2\pi} = \frac{2}{\pi},$$

TM (E-Polarized) Incidence (cont.)

so that dividing the echo width through by λ yields

$$\sigma_{bi}/\lambda = \frac{2}{\pi} \left| \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi} \right|^2. \quad (4.3.8)$$

The corresponding magnetic field should not be left implicit. Faraday's law $\vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H}$ and the cylindrical curl of $\hat{z}E_z$ give

$$H_\rho = \frac{i}{\omega\mu\rho} \partial_\phi E_z, \quad H_\phi = -\frac{i}{\omega\mu} \partial_\rho E_z. \quad (4.3.9)$$

Since $\omega\mu = k\eta$, applying (4.3.9) term by term to the incident field (4.3.1) yields the display below. Primes denote differentiation with respect to the complete dimensionless argument.

$$\begin{aligned} H_\rho^i &= -\frac{E_0}{\eta k \rho} \sum_{n=-\infty}^{\infty} n i^{-n} J_n(k\rho) e^{in\phi}, \\ H_\phi^i &= -\frac{iE_0}{\eta} \sum_{n=-\infty}^{\infty} i^{-n} J'_n(k\rho) e^{in\phi}. \end{aligned} \quad (4.3.10)$$

TM (E-Polarized) Incidence (cont.)

Applying the same derivatives to (4.3.5) gives the scattered magnetic field,

$$\begin{aligned} H_{\rho}^s &= \frac{E_0}{\eta k \rho} \sum_{n=-\infty}^{\infty} n i^{-n} \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(k\rho) e^{in\phi}, \\ H_{\phi}^s &= \frac{iE_0}{\eta} \sum_{n=-\infty}^{\infty} i^{-n} \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)'}(k\rho) e^{in\phi}. \end{aligned} \quad (4.3.11)$$

Adding (4.3.10) and (4.3.11), or directly differentiating (4.3.6), gives

$$\begin{aligned} H_{\rho}^t &= -\frac{E_0}{\eta k \rho} \sum_{n=-\infty}^{\infty} n i^{-n} \left[J_n(k\rho) - \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(k\rho) \right] e^{in\phi}, \\ H_{\phi}^t &= -\frac{iE_0}{\eta} \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n'(k\rho) - \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)'}(k\rho) \right] e^{in\phi}. \end{aligned} \quad (4.3.12)$$

TM (E-Polarized) Incidence (cont.)

The electric surface-current density on the PEC follows from $\vec{K} = \hat{\rho} \times \vec{H}_t|_{\rho=a}$. Only H_ϕ^t contributes, so

$$\begin{aligned}\vec{K}(\phi) &= \hat{z} H_\phi^t(a, \phi) \\ &= -\hat{z} \frac{iE_0}{\eta} \sum_{n=-\infty}^{\infty} i^{-n} e^{in\phi} \\ &\quad \times \left[J'_n(ka) - \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)'}(ka) \right].\end{aligned}\tag{4.3.13}$$

The ordinary-Bessel Wronskian is

$$J_n(x) H_n^{(2)'}(x) - J_n'(x) H_n^{(2)}(x) = -\frac{2i}{\pi x}.$$

It reduces the bracket according to

$$J_n'(x) - \frac{J_n(x)}{H_n^{(2)}(x)} H_n^{(2)'}(x) = \frac{2i}{\pi x H_n^{(2)}(x)}.$$

TM (E-Polarized) Incidence (cont.)

Consequently, an equivalent compact form is

$$\vec{K}(\phi) = \hat{z} \frac{2E_0}{\pi\eta ka} \sum_{n=-\infty}^{\infty} \frac{i^{-n} e^{in\phi}}{H_n^{(2)}(ka)}. \quad (4.3.14)$$

TE (H-Polarized) Incidence

For the complementary TE case, let the incident magnetic field point along the cylinder axis and define $H_0 = E_0/\eta$. Ampère–Maxwell's law in the source-free host, $\vec{\nabla} \times \vec{H} = i\omega\epsilon\vec{E}$, gives

$$\begin{aligned}\vec{H}_i &= \hat{z}H_0e^{-ikx}, \\ \vec{E}_i &= \frac{1}{i\omega\epsilon}\vec{\nabla} \times \vec{H}_i = \hat{y}\frac{k}{\omega\epsilon}H_0e^{-ikx} = \hat{y}E_0e^{-ikx},\end{aligned}\tag{4.3.15}$$

where $k/(\omega\epsilon) = \eta$. Expanding the incident field and choosing the outgoing Hankel function for the scattered field, write

$$H_z^i = H_0 \sum_{n=-\infty}^{\infty} i^{-n} J_n(k\rho) e^{in\phi},\tag{4.3.16}$$

$$H_z^s = H_0 \sum_{n=-\infty}^{\infty} i^{-n} q_n H_n^{(2)}(k\rho) e^{in\phi}.\tag{4.3.17}$$

TE (H-Polarized) Incidence (cont.)

For a field with only a z-directed magnetic component, the cylindrical curl reduces to

$$E_\rho = \frac{1}{i\omega\epsilon\rho} \partial_\phi H_z, \quad E_\phi = -\frac{1}{i\omega\epsilon} \partial_\rho H_z. \quad (4.3.18)$$

The PEC condition is continuity to zero of the tangential electric field at $\rho = a$. Substitution of (4.3.16) and (4.3.17) into (4.3.18) gives

$$\begin{aligned} E_\phi^t(a, \phi) &= -\frac{kH_0}{i\omega\epsilon} \sum_{n=-\infty}^{\infty} i^{-n} \left[J'_n(ka) + q_n H_n^{(2)'}(ka) \right] e^{in\phi} \\ &= iE_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J'_n(ka) + q_n H_n^{(2)'}(ka) \right] e^{in\phi} = 0. \end{aligned} \quad (4.3.19)$$

Fourier orthogonality forces each bracket in (4.3.19) to vanish separately, so

$$q_n = -\frac{J'_n(ka)}{H_n^{(2)'}(ka)}. \quad (4.3.20)$$

TE (H-Polarized) Incidence (cont.)

Thus the scattered and total axial magnetic fields are

$$H_z^s = -H_0 \sum_{n=-\infty}^{\infty} i^{-n} \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(k\rho) e^{in\phi}, \quad (4.3.21)$$

$$H_z^t = H_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n(k\rho) - \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(k\rho) \right] e^{in\phi}. \quad (4.3.22)$$

Applying (4.3.18) term by term makes the electric fields equally explicit. In particular,

$$\begin{aligned} E_\rho^s &= -\frac{E_0}{k\rho} \sum_{n=-\infty}^{\infty} n i^{-n} \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(k\rho) e^{in\phi}, \\ E_\phi^s &= -iE_0 \sum_{n=-\infty}^{\infty} i^{-n} \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)'}(k\rho) e^{in\phi}, \end{aligned} \quad (4.3.23)$$

TE (H-Polarized) Incidence (cont.)

and

$$\begin{aligned} E_{\rho}^t &= \frac{E_0}{k\rho} \sum_{n=-\infty}^{\infty} n i^{-n} \left[J_n(k\rho) - \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(k\rho) \right] e^{in\phi}, \\ E_{\phi}^t &= iE_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J'_n(k\rho) - \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)'}(k\rho) \right] e^{in\phi}. \end{aligned} \quad (4.3.24)$$

The second bracket in (4.3.24) vanishes identically at $\rho = a$, directly checking the PEC boundary condition. The induced current is now azimuthal because

$$\begin{aligned} \vec{K}(\phi) &= \hat{\rho} \times \hat{z} H_z^t(a, \phi) = -\hat{\phi} H_z^t(a, \phi) \\ &= -\hat{\phi} H_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n(ka) - \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(ka) \right] e^{in\phi}. \end{aligned} \quad (4.3.25)$$

TE (H-Polarized) Incidence (cont.)

In the far zone, (4.2.17) again cancels the factor i^{-n} against $e^{in\pi/2} = i^n$. Moreover, $H_n^{(2)'}(k\rho) \simeq -iH_n^{(2)}(k\rho)$, so the leading electric field is azimuthal and satisfies $E_\phi^s \simeq \eta H_z^s$. Therefore

$$\begin{aligned}\vec{E}_s &\simeq -\hat{\phi} E_0 \sqrt{\frac{2i}{\pi k}} \frac{e^{-ik\rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{\infty} \frac{J_n'(ka)}{H_n^{(2)'}(ka)} e^{in\phi}, \\ \vec{f}_{\text{TE}}(\phi) &= -\hat{\phi} \sqrt{\frac{2i}{\pi k}} \sum_{n=-\infty}^{\infty} \frac{J_n'(ka)}{H_n^{(2)'}(ka)} e^{in\phi}.\end{aligned}\tag{4.3.26}$$

The incident flux may be written either as $E_0^2/(2\eta)$ or as $\eta H_0^2/2$; hence the same 2D normalization used for the TM case applies. The TE echo width and its wavelength-normalized form are consequently

$$\sigma_{bi}^{\text{TE}} = \frac{4}{k} \left| \sum_{n=-\infty}^{\infty} \frac{J_n'(ka)}{H_n^{(2)'}(ka)} e^{in\phi} \right|^2,\tag{4.3.27}$$

$$\frac{\sigma_{bi}^{\text{TE}}}{\lambda} = \frac{2}{\pi} \left| \sum_{n=-\infty}^{\infty} \frac{J_n'(ka)}{H_n^{(2)'}(ka)} e^{in\phi} \right|^2.\tag{4.3.28}$$

Table of contents

- 1 Cross Sections
- 2 Cylindrical Waves
 - Cylindrical Wave Solution
 - Cylindrical Wave Transformation
- 3 Scattering from PEC Cylinders
 - TM (E-Polarized) Incidence
 - TE (H-Polarized) Incidence
- 4 Spherical Waves
 - Spherical Wave Solution
 - Spherical Wave Transformation
- 5 Scattering from Dielectric Spheres

Spherical Waves

This section treats wave solutions in spherical coordinates; unlike the cylindrical case, these require the special considerations developed below.

Spherical Wave Solution

The application of the vector Helmholtz equation in spherical coordinates is not straightforward, as none of the unit vectors are constant throughout space. As a result, the vector Helmholtz equation cannot be directly reduced to separate scalar Helmholtz equations for each component. To illustrate this, consider (3.1.4)

$\vec{\nabla} \times \vec{\nabla} \times \vec{A} - k^2 \vec{A} = \mu \vec{J} - i\omega\mu\epsilon \vec{\nabla}\varphi$ in spherical coordinates. By assuming that $\vec{A}$ and $\vec{J}$ have only radial component, that is

$$\vec{A} = A_r \hat{r}, \quad \vec{J} = J_r \hat{r}. \quad (4.4.1)$$

Then, the r , θ , and ϕ components of (3.1.4) are given by

$$\left[\frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 + k^2 \right] A_r = i\omega\mu\epsilon \partial_r \varphi - \mu J_r, \quad (4.4.2)$$

$$-\frac{1}{r} \partial_r \partial_\theta A_r = \frac{i\omega\mu\epsilon}{r} \partial_\theta \varphi, \quad (4.4.3)$$

$$-\frac{1}{r \sin \theta} \partial_r \partial_\phi A_r = \frac{i\omega\mu\epsilon}{r \sin \theta} \partial_\phi \varphi. \quad (4.4.4)$$

Spherical Wave Solution (cont.)

These follow from evaluating $\vec{B} = \vec{\nabla} \times \vec{A}$ for $\vec{A} = A_r \hat{r}$. With $A_\theta = A_\phi = 0$, the spherical curl gives

$$B_r = 0, \quad B_\theta = \frac{1}{r \sin \theta} \partial_\phi A_r, \quad B_\phi = -\frac{1}{r} \partial_\theta A_r.$$

Taking the curl once more, the radial component is

$$\begin{aligned} & \frac{1}{r \sin \theta} [\partial_\theta (\sin \theta B_\phi) - \partial_\phi B_\theta] \\ &= - \left[\frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 \right] A_r, \end{aligned}$$

which carries no $\partial_r^2 A_r$ term because $B_r = 0$, while the transverse components are

$$(\vec{\nabla} \times \vec{B})_\theta = \frac{1}{r} \partial_r \partial_\theta A_r, \quad (\vec{\nabla} \times \vec{B})_\phi = \frac{1}{r \sin \theta} \partial_r \partial_\phi A_r.$$

Equating $\vec{\nabla} \times \vec{\nabla} \times \vec{A} - k^2 \vec{A}$ component by component to the right-hand side $\mu \vec{J} - i\omega \mu \epsilon \vec{\nabla} \varphi$ of (3.1.4) (using $(\vec{\nabla} \varphi)_\theta = \frac{1}{r} \partial_\theta \varphi$ and $(\vec{\nabla} \varphi)_\phi = \frac{1}{r \sin \theta} \partial_\phi \varphi$) reproduces

Spherical Wave Solution (cont.)

(4.4.2)–(4.4.4). We notice that the Lorenz gauge condition (3.1.8) cannot be used to simplify (4.4.3) and (4.4.4); instead, we choose

$$\partial_r A_r + i\omega\mu\epsilon\varphi = 0. \quad (4.4.5)$$

By doing so, (4.4.3) and (4.4.4) are satisfied. Indeed, differentiating the gauge condition (4.4.5) with respect to θ and ϕ (which commute with ∂_r) gives

$$i\omega\mu\epsilon\partial_\theta\varphi = -\partial_r\partial_\theta A_r, \quad i\omega\mu\epsilon\partial_\phi\varphi = -\partial_r\partial_\phi A_r.$$

Substituting these into the right-hand sides of (4.4.3) and (4.4.4) reproduces their left-hand sides exactly, e.g.

$$\frac{i\omega\mu\epsilon}{r}\partial_\theta\varphi = -\frac{1}{r}\partial_r\partial_\theta A_r,$$

so both reduce to the identity $0 = 0$. Likewise, differentiating (4.4.5) with respect to r gives

$$i\omega\mu\epsilon\partial_r\varphi = -\partial_r^2 A_r,$$

Spherical Wave Solution (cont.)

which is the scalar-potential term on the right-hand side of (4.4.2). Moving it to the left-hand side supplies the missing radial second derivative $\partial_r^2 A_r$ and leaves only $-\mu J_r$ on the right, so that

$$\left[\partial_r^2 + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 + k^2 \right] A_r = -\mu J_r. \quad (4.4.6)$$

Let us introduce the Debye potential:

$$\pi_e = \frac{A_r}{i\omega\mu\epsilon r}. \quad (4.4.7)$$

Then (4.4.6) is transformed into the scalar Helmholtz equation in spherical coordinates with $\nabla^2 = \frac{1}{r^2} \partial_r (r^2 \partial_r) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2$. To see this, write (4.4.7) as $A_r = i\omega\mu\epsilon r \pi_e$ and substitute it into (4.4.6). Because r is constant with respect to θ and ϕ , the two transverse terms simply pass the factor $i\omega\mu\epsilon r$ through, while the radial term uses the identity

$$\partial_r^2 (r \pi_e) = r \left[\frac{1}{r^2} \partial_r (r^2 \partial_r \pi_e) \right],$$

Spherical Wave Solution (cont.)

which follows from $\partial_r^2(r\pi_e) = \partial_r(\pi_e + r\partial_r\pi_e) = 2\partial_r\pi_e + r\partial_r^2\pi_e$ and $\frac{1}{r^2}\partial_r(r^2\partial_r\pi_e) = \frac{2}{r}\partial_r\pi_e + \partial_r^2\pi_e$. Hence every term on the left of (4.4.6) carries the common factor $i\omega\mu\epsilon r$,

$$\begin{aligned} & \left[\partial_r^2 + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 + k^2 \right] A_r \\ &= i\omega\mu\epsilon r \left[\frac{1}{r^2} \partial_r (r^2 \partial_r) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) \right. \\ & \quad \left. + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 + k^2 \right] \pi_e \\ &= i\omega\mu\epsilon r (\nabla^2 + k^2) \pi_e. \end{aligned}$$

Equating this to the right-hand side $-\mu J_r$ of (4.4.6) and dividing through by $i\omega\mu\epsilon r$ yields, since $-\mu J_r/(i\omega\mu\epsilon r) = -J_r/(i\omega\epsilon r)$,

$$(\nabla^2 + k^2) \pi_e = -\frac{J_r}{i\omega\epsilon r}. \quad (4.4.8)$$

Spherical Wave Solution (cont.)

By duality, the radial magnetic current density $\vec{M} = M_r \hat{r}$ similarly gives

$$(\nabla^2 + k^2) \pi_m = -\frac{M_r}{i\omega\mu r}. \quad (4.4.9)$$

Once the solutions of (4.4.8) and (4.4.9) are found, the fields can be found by plugging (4.4.5), (4.4.7) into (3.1.3), (3.1.1) and applying the duality transform. Before these quantities are first used, define the scalar radial amplitudes and their corresponding radial vectors by

$$\Pi_e \equiv r\pi_e, \quad \vec{\Pi}_e \equiv \Pi_e \hat{r}, \quad \Pi_m \equiv r\pi_m, \quad \vec{\Pi}_m \equiv \Pi_m \hat{r}.$$

Let us first carry out the electric part explicitly. The gauge choice (4.4.5) fixes the scalar potential as $\varphi = -\frac{1}{i\omega\mu\epsilon} \partial_r A_r$, while (4.4.7) gives $A_r = i\omega\mu\epsilon r\pi_e = i\omega\mu\epsilon \Pi_e$. Substituting both into (3.1.3),

$$\begin{aligned} \vec{E} &= -\vec{\nabla}\varphi - i\omega\vec{A} \\ &= \frac{1}{i\omega\mu\epsilon} \vec{\nabla}(\partial_r A_r) - i\omega A_r \hat{r} \\ &= \vec{\nabla}(\partial_r \Pi_e) + \omega^2 \mu\epsilon \Pi_e \hat{r}, \end{aligned}$$

Spherical Wave Solution (cont.)

which, using $k^2 = \omega^2 \mu \epsilon$, is exactly the electric part $\vec{\nabla} \partial_r \Pi_e + k^2 \vec{\Pi}_e$. The magnetic contribution $-i\omega\mu \vec{\nabla} \times \vec{\Pi}_m$ is then appended by duality, yielding (4.4.12) below; the $\vec{H}$ expression follows from $\Pi_e \rightarrow \Pi_m$, $\Pi_m \rightarrow -\Pi_e$, $\mu \rightarrow \epsilon$.

To obtain the components, recall the spherical gradient

$$\vec{\nabla} f = \hat{r} \partial_r f + \hat{\theta} \frac{1}{r} \partial_\theta f + \hat{\phi} \frac{1}{r \sin \theta} \partial_\phi f,$$

and the curl of a purely radial field $\vec{\Pi}_m = \Pi_m \hat{r}$,

$$\vec{\nabla} \times \vec{\Pi}_m = \hat{\theta} \frac{1}{r \sin \theta} \partial_\phi \Pi_m - \hat{\phi} \frac{1}{r} \partial_\theta \Pi_m.$$

Applying the gradient to the scalar $\partial_r \Pi_e$ and adding the $k^2 \Pi_e \hat{r}$ term gives the electric part of each component, while $-i\omega\mu \vec{\nabla} \times \vec{\Pi}_m$ supplies the magnetic part; the $\hat{r}$ -curl vanishes, so E_r has no magnetic contribution.

$$\vec{E} = \vec{\nabla} \partial_r \Pi_e + k^2 \vec{\Pi}_e - i\omega\mu \vec{\nabla} \times \vec{\Pi}_m, \quad (4.4.10)$$

$$\vec{H} = \vec{\nabla} \partial_r \Pi_m + k^2 \vec{\Pi}_m + i\omega\epsilon \vec{\nabla} \times \vec{\Pi}_e. \quad (4.4.11)$$

Spherical Wave Solution (cont.)

The source-free Maxwell curls provide an independent sign check on this reconstruction. If $\pi = \Pi/r$ obeys $(\nabla^2 + k^2)\pi = 0$, multiplication by r gives

$$\partial_r^2 \Pi + \frac{1}{r^2} \nabla_a^2 \Pi + k^2 \Pi = 0,$$

where ∇_a^2 is the angular Laplacian. The curl-curl components derived above then give

$$\vec{\nabla} \times \vec{\nabla} \times (\Pi \hat{r}) = \vec{\nabla} \partial_r \Pi + k^2 \Pi \hat{r}.$$

Thus the electric-potential fields may be written $\vec{E}_e = \vec{\nabla} \times \vec{\nabla} \times \vec{\Pi}_e$ and $\vec{H}_e = i\omega\epsilon \vec{\nabla} \times \vec{\Pi}_e$. Their curls are

$$\vec{\nabla} \times \vec{E}_e = k^2 \vec{\nabla} \times \vec{\Pi}_e = -i\omega\mu \vec{H}_e,$$

$$\vec{\nabla} \times \vec{H}_e = i\omega\epsilon \vec{\nabla} \times \vec{\nabla} \times \vec{\Pi}_e = i\omega\epsilon \vec{E}_e,$$

because $k^2 = \omega^2\mu\epsilon$. For the magnetic potential, $\vec{H}_m = \vec{\nabla} \times \vec{\nabla} \times \vec{\Pi}_m$ and $\vec{E}_m = -i\omega\mu \vec{\nabla} \times \vec{\Pi}_m$ give the same two Maxwell equations. This verifies both curl-term signs in the combined fields above.

Spherical Wave Solution (cont.)

In terms of spherical components, the electric field is expressed as:

$$E_r = (\partial_r^2 + k^2) \Pi_e, \quad (4.4.12a)$$

$$E_\theta = \frac{1}{r} \partial_r \partial_\theta \Pi_e - \frac{i\omega\mu}{r \sin \theta} \partial_\phi \Pi_m, \quad (4.4.12b)$$

$$E_\phi = \frac{1}{r \sin \theta} \partial_r \partial_\phi \Pi_e + \frac{i\omega\mu}{r} \partial_\theta \Pi_m, \quad (4.4.12c)$$

and the magnetic field is expressed as:

$$H_r = (\partial_r^2 + k^2) \Pi_m, \quad (4.4.13a)$$

$$H_\theta = \frac{1}{r} \partial_r \partial_\theta \Pi_m + \frac{i\omega\epsilon}{r \sin \theta} \partial_\phi \Pi_e, \quad (4.4.13b)$$

$$H_\phi = \frac{1}{r \sin \theta} \partial_r \partial_\phi \Pi_m - \frac{i\omega\epsilon}{r} \partial_\theta \Pi_e. \quad (4.4.13c)$$

We derived the source terms in (4.4.8)–(4.4.9) only for radial electric and magnetic currents. Transverse source components require a separate projection and cannot simply

Spherical Wave Solution (cont.)

be inserted into those right-hand sides. In any homogeneous region away from all sources, the right-hand sides are set to zero.

The completeness of the two homogeneous potentials can be seen mode by mode. Let $Y_{nm}(\theta, \phi) \propto P_n^m(\cos \theta)e^{im\phi}$ denote a scalar spherical harmonic. The three vector spherical harmonics

$$Y_{nm}\hat{r}, \quad \nabla_a Y_{nm}, \quad \hat{r} \times \nabla_a Y_{nm}$$

form a complete angular basis for square-integrable vector fields on a sphere; here ∇_a is the dimensionless angular gradient on the unit sphere. For a sufficiently regular source-free Maxwell field in a homogeneous spherical ball or shell centered at the chosen origin, a separated electric potential $\pi_e = R_n(r)Y_{nm}$, with R_n the radial function introduced below, generates an electric field in the first two directions and, through $\vec{H}_e = i\omega\epsilon\vec{\nabla} \times (\Pi_e\hat{r})$, a magnetic field in the third direction. This is the TM-to- r spherical mode. Conversely, $\pi_m = R_n(r)Y_{nm}$ generates the same pattern with $\vec{E}$ and $\vec{H}$ interchanged, giving the TE-to- r mode. For each n, m , the radial equation has two independent solutions, so these two polarizations with their allowed radial solutions span every mode in that source-free field class.

There is no missing dynamic $n = 0$ mode. For $n = 0$ the angular derivatives vanish, while the radial equation gives $(\partial_r^2 + k^2)[rR_0(kr)] = 0$; hence every field component

Spherical Wave Solution (cont.)

reconstructed above is zero. Equivalently, a nonzero time-harmonic electromagnetic monopole would contradict the source-free Maxwell curl equations. Thus the physical sum begins at $n = 1$, and the two scalar Debye potentials form a complete representation under the regularity and spherical-region hypotheses just stated. When $\Pi_m = 0$ only the TM family remains; when $\Pi_e = 0$ only the TE family remains.

Now consider the scalar Helmholtz equation in spherical coordinates

$$\frac{1}{r^2} \partial_r (r^2 \partial_r \psi) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta \psi) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 \psi + k^2 \psi = 0. \quad (4.4.14)$$

Using separation of variables, let

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi). \quad (4.4.15)$$

Plug into (4.4.14), multiply $r^2 \sin^2 \theta$ and divide it by ψ , we get

$$\begin{aligned} \frac{\sin^2 \theta}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} \\ + (kr \sin \theta)^2 = 0. \end{aligned} \quad (4.4.16)$$

Spherical Wave Solution (cont.)

Similar to the cylindrical case, for the Φ term, we have

$$\frac{d^2\Phi}{d\phi^2} + \nu^2\Phi = 0, \quad \Phi = e^{\pm i\nu\phi}, \quad (4.4.17)$$

where ν is a constant. Plug into (4.4.16) and divide it by $\sin^2\theta$, we get

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{\Theta \sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) - \frac{\nu^2}{\sin^2\theta} + (kr)^2 = 0. \quad (4.4.18)$$

Now, introduce a dimensionless separation constant λ (distinct from the wavelength $\lambda = 2\pi/k$ used earlier) via

$$\frac{1}{\Theta \sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) - \frac{\nu^2}{\sin^2\theta} = -\lambda, \quad (4.4.19)$$

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + (kr)^2 = \lambda. \quad (4.4.20)$$

Since Φ is periodic over 2π , we let $\nu = m$ an integer value, and from (4.4.19), we get

$$\frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) + \left(\lambda - \frac{m^2}{\sin^2\theta} \right) \Theta = 0. \quad (4.4.21)$$

Spherical Wave Solution (cont.)

Let us consider two cases: the solution is azimuthal symmetric or non-azimuthal symmetric. If it is azimuthal symmetric, we have $m = 0$ and (4.4.21) is reduced to

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \lambda \Theta = 0. \quad (4.4.22)$$

For a nonzero angular eigenfunction to remain finite at both poles, the dimensionless separation eigenvalue must be $\lambda = n(n+1)$, where n is a nonnegative integer. This restriction is not arbitrary. Introducing the Legendre argument $x = \cos \theta$ (distinct from the Mie size parameter $x = ka$ used later), so that $\sin^2 \theta = 1 - x^2$ and $\sin \theta d\theta = -dx$, the azimuthally symmetric equation (4.4.22) takes the self-adjoint form

$$\frac{d}{dx} \left[(1 - x^2) \frac{d\Theta}{dx} \right] + \lambda \Theta = 0,$$

valid on $-1 \leq x \leq 1$. The coefficient $1 - x^2$ vanishes at $x = \pm 1$ (i.e. at $\cos \theta = \pm 1$), so these endpoints are regular singular points. The point $x = 0$ is ordinary; seeking a Taylor series $\Theta = \sum_{q \geq 0} a_q x^q$ there and matching like powers gives the two-term recurrence

$$a_{q+2} = \frac{q(q+1) - \lambda}{(q+1)(q+2)} a_q.$$

Spherical Wave Solution (cont.)

The recurrence separates into even and odd coefficient chains. For generic λ they do not terminate; a suitable linear combination can be regular at one endpoint, but the second endpoint then contains the logarithmically singular Legendre solution. Regularity at both endpoints is obtained when one chain terminates, which requires its numerator to vanish at some integer $q = n$:

$$\lambda = n(n+1), \quad n = 0, 1, 2, \dots,$$

in which case the series collapses to the Legendre polynomial $P_n(x) = P_n(\cos \theta)$ and the second, unbounded solution is discarded. Thus, we arrive at the following equation

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + n(n+1)\Theta = 0, \quad (4.4.23)$$

which is known as the Legendre equation, and their solutions are known as the Legendre polynomials:

$$P_n(\cos \theta) = \frac{1}{2^n n!} \left(\frac{d}{d \cos \theta} \right)^n (\cos^2 \theta - 1)^n. \quad (4.4.24)$$

Spherical Wave Solution (cont.)

The Legendre polynomials form a complete orthogonal basis of $L^2([-1, 1])$. Their orthogonality relation is

$$\int_{-1}^1 P_n(\cos \theta) P_{n'}(\cos \theta) d \cos \theta = \frac{2}{2n+1} \delta_{nn'}. \quad (4.4.25)$$

Hence, if $g(x) = f(\arccos x)$ belongs to $L^2([-1, 1])$, its Legendre series is

$$f(\theta) = \sum_{n=0}^{\infty} a_n P_n(\cos \theta) \quad \text{in } L^2([0, \pi], \sin \theta d\theta), \quad (4.4.26)$$

where

$$a_n = \frac{2n+1}{2} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta. \quad (4.4.27)$$

The L^2 statement does not by itself assert pointwise convergence; such a claim requires additional regularity of f and must be checked under the appropriate Legendre-series

Spherical Wave Solution (cont.)

convergence theorem. On the other hand, if the solution is not azimuthally symmetric, we obtain the associated Legendre equation

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \left[n(n+1) - \frac{m^2}{\sin^2 \theta} \right] \Theta = 0. \quad (4.4.28)$$

The solutions to (4.4.28) are the associated Legendre functions of the first kind $P_n^m(\cos \theta)$ and second kind $Q_n^m(\cos \theta)$. Here, m is the integer azimuthal order introduced above. Requiring one angular eigenfunction to remain bounded at *both* poles selects integer $n \geq 0$ and integer m with $|m| \leq n$; the bounded first-kind solution is $P_n^m(\cos \theta)$, while the second-kind solution is singular at a pole. The real azimuthal basis has only the constant cosine mode $\cos 0\phi = 1$ at $m = 0$; for every $m = 1, \dots, n$ it has the independent pair $\cos m\phi$ and $\sin m\phi$. Throughout this book we use the phase-free engineering convention for the associated Legendre functions,

$$P_n^m(\cos \theta) = \sin^m \theta P_n^{(m)}(\cos \theta), \quad 0 \leq m \leq n, \quad (4.4.29)$$

where $P_n^{(m)}$ denotes the m^{th} derivative of the n^{th} -order Legendre polynomial with respect to $\cos \theta$. Thus our convention omits the Condon–Shortley factor $(-1)^m$; formulas imported from a source that includes that phase must be multiplied by $(-1)^m$.

Spherical Wave Solution (cont.)

Now let us consider (4.4.20). Substituting $\lambda = n(n+1)$ and letting

$$R = R_n(kr) = \sqrt{\frac{\pi}{2kr}} B_{n+1/2}(kr) = \sqrt{\frac{\pi}{2kr}} B, \quad (4.4.30)$$

we first note that (4.4.20) may be written in the product form

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + [(kr)^2 - n(n+1)] R = 0.$$

Writing $R = \sqrt{\pi/2k} r^{-1/2} B$ from (4.4.30), the radial derivative is

$$r^2 \frac{dR}{dr} = \sqrt{\frac{\pi}{2k}} \left[r^{3/2} \frac{dB}{dr} - \frac{1}{2} r^{1/2} B \right],$$

so that

$$\begin{aligned} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) &= \sqrt{\frac{\pi}{2k}} \left[r^{3/2} \frac{d^2 B}{dr^2} + r^{1/2} \frac{dB}{dr} \right. \\ &\quad \left. - \frac{1}{4} r^{-1/2} B \right]. \end{aligned}$$

Spherical Wave Solution (cont.)

Adding $[(kr)^2 - n(n+1)]R = \sqrt{\pi/2k} r^{-1/2} [(kr)^2 - n(n+1)]B$ and multiplying through by $r^{-3/2} \sqrt{2k/\pi}$ gives

$$\frac{d^2 B}{dr^2} + \frac{1}{r} \frac{dB}{dr} + \left[k^2 - \frac{n(n+1) + \frac{1}{4}}{r^2} \right] B = 0.$$

Since $n(n+1) + \frac{1}{4} = (n + \frac{1}{2})^2$, this is

$$\frac{d^2 B}{dr^2} + \frac{1}{r} \frac{dB}{dr} + \left[k^2 - \frac{(n + 1/2)^2}{r^2} \right] B = 0, \quad (4.4.31)$$

which is the Bessel equation and $B_{n+1/2}$ is the Bessel function of order $n + 1/2$. (4.4.30) is known as the spherical Bessel function. We often use lowercase letters to denote the spherical forms of the Bessel and Hankel functions. For example, the spherical Bessel and Hankel functions are defined as

$$j_n(kr) = \sqrt{\frac{\pi}{2kr}} J_{n+1/2}(kr), \quad h_n^{(2)}(kr) = \sqrt{\frac{\pi}{2kr}} H_{n+1/2}^{(2)}(kr).$$

Spherical Wave Transformation

Similar to (4.2.18), consider an x-polarized plane wave propagating in the +z-direction:

$$\vec{E} = \hat{x} E_0 e^{-ikz} = \hat{x} E_0 e^{-ikr \cos \theta}. \quad (4.4.32)$$

Let us expand the exponential term into an infinite series of spherical waves. The *scalar phase factor* $e^{-ikr \cos \theta}$ is azimuthally symmetric, so its scalar expansion has $m = 0$; the fixed Cartesian polarization vector $\hat{x}$ itself is not azimuthally symmetric when resolved into spherical components. Furthermore, because the solution must remain finite at the origin, we have

$$e^{-ikr \cos \theta} = \sum_{n=0}^{\infty} p_n^{(\text{sph})} j_n(kr) P_n(\cos \theta). \quad (4.4.33)$$

The superscript distinguishes these temporary scalar plane-wave coefficients from the Mie scattering coefficients introduced below. To find $p_n^{(\text{sph})}$, use the orthogonality relation (4.4.25):

$$p_n^{(\text{sph})} j_n(kr) = \frac{2n+1}{2} \int_{-1}^1 e^{-ikr \cos \theta} P_n(\cos \theta) d \cos \theta. \quad (4.4.34)$$

Spherical Wave Transformation (cont.)

The required integral identity can be derived without skipping the projection step. Put $u = \cos \theta$ and $z = kr$. Rodrigues' formula and n integrations by parts give

$$\begin{aligned} I_n(z) &= \int_{-1}^1 e^{-izu} P_n(u) du \\ &= \frac{1}{2^n n!} \int_{-1}^1 e^{-izu} \frac{d^n}{du^n} (u^2 - 1)^n du \\ &= \frac{(-i)^n z^n}{2^n n!} \int_{-1}^1 e^{-izu} (1 - u^2)^n du. \end{aligned}$$

All boundary terms vanish because $(u^2 - 1)^n$ and its first $n - 1$ derivatives vanish at $u = \pm 1$. The integral representation

$$j_n(z) = \frac{z^n}{2^{n+1} n!} \int_{-1}^1 e^{-izu} (1 - u^2)^n du$$

then yields

$$\int_{-1}^1 e^{-ikr \cos \theta} P_n(\cos \theta) d \cos \theta = 2i^{-n} j_n(kr), \quad (4.4.35)$$

Spherical Wave Transformation (cont.)

because $(-i)^n = i^{-n}$. For fixed n , $j_n(kr)$ is not identically zero, so the same cancellation argument used above gives a coefficient independent of r :

$$p_n^{(\text{sph})} = (2n+1)i^{-n}. \quad (4.4.36)$$

Thus, we have the following expansion

$$\begin{aligned} E_x &= E_0 e^{-ikz} = E_0 e^{-ikr \cos \theta} \\ &= E_0 \sum_{n=0}^{\infty} (2n+1) i^{-n} j_n(kr) P_n(\cos \theta). \end{aligned} \quad (4.4.37)$$

This is called the spherical wave transformation which expands a plane wave to a sum of spherical waves.

Table of contents

- 1 Cross Sections
- 2 Cylindrical Waves
 - Cylindrical Wave Solution
 - Cylindrical Wave Transformation
- 3 Scattering from PEC Cylinders
 - TM (E-Polarized) Incidence
 - TE (H-Polarized) Incidence
- 4 Spherical Waves
 - Spherical Wave Solution
 - Spherical Wave Transformation
- 5 Scattering from Dielectric Spheres

Scattering from Dielectric Spheres

The exact solution for the scattering of a plane electromagnetic wave by an isotropic, homogeneous dielectric sphere of arbitrary size is commonly known as Mie theory.

Consider a dielectric sphere of radius a , permittivity ϵ_d , and permeability μ_d , placed in a host medium with permittivity ϵ and permeability μ . We use

$$\begin{aligned} k &= \omega\sqrt{\epsilon\mu}, & \eta &= \frac{\omega\mu}{k} = \frac{k}{\omega\epsilon}, \\ k_d &= \omega\sqrt{\epsilon_d\mu_d}, & \eta_d &= \frac{\omega\mu_d}{k_d} = \frac{k_d}{\omega\epsilon_d}. \end{aligned} \tag{4.5.1}$$

For the lossless host, k and η are chosen positive. All interior square roots are taken consistently with k_d ; with the $e^{i\omega t}$ convention, a passive lossy material has the branch $\text{Im } k_d \leq 0$. The normalization and divided coefficient formulas below assume $\epsilon_d \neq 0$, $\mu_d \neq 0$, and hence $k_d \neq 0$. Epsilon-near-zero, mu-near-zero, and exact zero-index cases must be obtained by returning to the undivided tangential-field matching equations and taking the appropriate material limit; direct substitution into expressions containing $1/k_d$ or $1/\eta_d$ is not defined. Let the incident electric field be an x -polarized plane wave

Scattering from Dielectric Spheres (cont.)

propagating in the $+z$ -direction. The radial component of the incident electric field can be expressed as

$$\begin{aligned} E_r^i &= E_0 e^{-ikz} \hat{x} \cdot \hat{r} = E_0 \sin \theta \cos \phi e^{-ikr \cos \theta} \\ &= \frac{1}{ikr} E_0 \cos \phi \partial_\theta e^{-ikr \cos \theta}. \end{aligned} \quad (4.5.2)$$

Expanding the phase factor via (4.4.37) then gives

$$\begin{aligned} E_r^i &= \frac{E_0 \cos \phi}{ikr} \partial_\theta \sum_{n=0}^{\infty} (2n+1) i^{-n} j_n(kr) P_n(\cos \theta) \\ &= \frac{iE_0 \cos \phi}{(kr)^2} \sum_{n=1}^{\infty} (2n+1) i^{-n} \hat{j}_n(kr) P_n^1(\cos \theta), \end{aligned} \quad (4.5.3)$$

where

$$-\partial_\theta P_n(\cos \theta) = P_n^1(\cos \theta), \quad (4.5.4)$$

Scattering from Dielectric Spheres (cont.)

$$\hat{j}_n(kr) = krj_n(kr). \quad (4.5.5)$$

To find the electric Debye potential of the incident field, (4.4.12a) must be satisfied:

$$E_r^i = (\partial_r^2 + k^2) \Pi_e^i = (\partial_r^2 + k^2) (r\pi_e^i). \quad (4.5.6)$$

Let us expand the Debye potential in terms of spherical harmonics

$$\pi_e^i = \sum_{n=0}^{\infty} \sum_{m=0}^n j_n(kr) P_n^m(\cos \theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi). \quad (4.5.7)$$

Here and in every uniform real-basis double sum, $B_{0n} \equiv 0$ rather than an independent unknown: the $m = 0$ contribution is solely $A_{0n}P_n(\cos \theta)$ because $\sin 0\phi = 0$, so the sine-cosine pair begins at $m = 1$. Substituting (4.5.7) into (4.5.6), we get

$$E_r^i = \sum_{n=0}^{\infty} \sum_{m=0}^n \frac{n(n+1)}{kr^2} \hat{j}_n(kr) P_n^m(\cos \theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi). \quad (4.5.8)$$

Scattering from Dielectric Spheres (cont.)

Here, the standard identity used in the derivation of (4.5.8) is established as follows: since the operator $\partial_r^2 + k^2$ in (4.5.6) acts only on the radial factor, applying it term by term to (4.5.7) reduces the problem to evaluating $(\partial_r^2 + k^2)[r j_n(kr)]$. With $\lambda = n(n+1)$ and $R = j_n(kr)$, the radial equation (4.4.20) reads

$$\frac{d}{dr} \left(r^2 \frac{dj_n}{dr} \right) + [(kr)^2 - n(n+1)] j_n = 0,$$

which, after expanding the first term and dividing by r , becomes

$$r j_n'' + 2 j_n' + \left[k^2 r - \frac{n(n+1)}{r} \right] j_n = 0.$$

On the other hand, differentiating $r j_n(kr)$ twice gives

$$\begin{aligned} (\partial_r^2 + k^2)[r j_n(kr)] &= 2 j_n' + r j_n'' + k^2 r j_n \\ &= \frac{n(n+1)}{r} j_n(kr), \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

where the last equality follows directly from (4.4.20). Comparing (4.5.8) and (4.5.3), the right-hand side of (4.5.3) carries the single azimuthal harmonic $\cos \phi$ and the single associated-Legendre order P_n^1 , whereas (4.5.8) contains the full real basis $\{1\} \cup \{\cos m\phi, \sin m\phi\}_{m \geq 1}$. These nonzero basis functions are mutually orthogonal over $[0, 2\pi]$, so every harmonic that is absent on the left must have a vanishing coefficient on the right. Hence only the $m = 1$ cosine term survives,

$$A_{mn} = 0 \quad (m \neq 1), \quad B_{mn} = 0.$$

Equating the remaining n -th coefficients of $\cos \phi P_n^1 \hat{j}_n(kr)$ in (4.5.8) and (4.5.3) gives

$$\frac{n(n+1)}{kr^2} A_{1n} = \frac{i(2n+1)i^{-n}}{(kr)^2} E_0.$$

Scattering from Dielectric Spheres (cont.)

By the same cancellation argument used above— $\hat{j}_n(kr)$, $\cos \phi$, and P_n^1 are nonzero away from isolated points—the common factors, including $1/r^2$, cancel to give a coefficient identity independent of r . One extra power of k remains on the right, so

$$A_{1n} = E_0 i^{1-n} \frac{2n+1}{kn(n+1)}.$$

Writing the phase as $i^{1-n} = (i^{-1})^{n-1} = (-i)^{n-1}$, this is

$$\begin{cases} A_{mn} = 0, & m \neq 1, \\ A_{1n} = E_0 (-i)^{n-1} \frac{2n+1}{kn(n+1)}, \end{cases} \quad (4.5.9)$$

$$B_{mn} = 0. \quad (4.5.10)$$

Scattering from Dielectric Spheres (cont.)

Thus, we have

$$\pi_e^i = \frac{E_0 \cos \phi}{k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} \hat{j}_n(kr) P_n^1(\cos \theta). \quad (4.5.11)$$

The magnetic potential follows without skipping the corresponding projection. Faraday's law for the incident plane wave gives

$$\vec{H}_i = \frac{1}{\eta} \hat{z} \times \vec{E}_i = \hat{y} \frac{E_0}{\eta} e^{-ikz},$$

and hence

$$\begin{aligned} H_r^i &= \vec{H}_i \cdot \hat{r} = \frac{E_0}{\eta} \sin \theta \sin \phi e^{-ikr \cos \theta} \\ &= \frac{E_0 \sin \phi}{i\eta kr} \partial_\theta e^{-ikr \cos \theta}. \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

Using the scalar plane-wave expansion (4.4.37), the phase-free identity $-\partial_\theta P_n(\cos \theta) = P_n^1(\cos \theta)$, and $j_n(kr) = \hat{j}_n(kr)/(kr)$ gives

$$H_r^i = \frac{iE_0 \sin \phi}{\eta(kr)^2} \sum_{n=1}^{\infty} (2n+1) i^{-n} \hat{j}_n(kr) P_n^1(\cos \theta).$$

Equating this term by term to $H_r^i = (\partial_r^2 + k^2)(r\pi_m^i)$ repeats the radial identity (4.5.8) with $E_0 \cos \phi$ replaced by $E_0 \sin \phi / \eta$. Therefore

$$\pi_m^i = \frac{E_0 \sin \phi}{\eta k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} \hat{j}_n(kr) P_n^1(\cos \theta). \quad (4.5.12)$$

To find the scattered field inside and outside the sphere, let us first specify the matching conditions to be applied at $r = a$:

$$E_\theta = E_\theta^d, \quad E_\phi = E_\phi^d, \quad (4.5.13)$$

Scattering from Dielectric Spheres (cont.)

$$H_\theta = H_\theta^d, \quad H_\phi = H_\phi^d, \quad (4.5.14)$$

where the superscript d denotes the field inside the sphere. From (4.4.12) and (4.4.13), the matching conditions involve both $\pi_e = \pi_e^i + \pi_e^s$ and $\pi_m = \pi_m^i + \pi_m^s$, making them coupled. To simplify the analysis, it is convenient to decouple them by deriving matching conditions for π_e and π_m individually. To achieve this, we consider a linear combination of E_θ and E_ϕ chosen so that all terms involving π_m cancel; applying a similar technique to the magnetic components then gives, at $r = a$:

$$\frac{1}{r} \partial_r (r \pi_e) = \frac{1}{r} \partial_r (r \pi_e^d), \quad \epsilon \pi_e = \epsilon_d \pi_e^d, \quad (4.5.15)$$

$$\frac{1}{r} \partial_r (r \pi_m) = \frac{1}{r} \partial_r (r \pi_m^d), \quad \mu \pi_m = \mu_d \pi_m^d. \quad (4.5.16)$$

Scattering from Dielectric Spheres (cont.)

To justify these conditions, substitute the field components (4.4.12) into the two independent angular combinations of the tangential field. Writing $\Pi_e = r\pi_e$ and $\Pi_m = r\pi_m$, the divergence-type combination of E_θ and E_ϕ gives

$$\begin{aligned}\partial_\theta(\sin\theta E_\theta) + \partial_\phi E_\phi \\&= \frac{1}{r} \partial_r \left[\partial_\theta(\sin\theta \partial_\theta \Pi_e) + \frac{1}{\sin\theta} \partial_\phi^2 \Pi_e \right] \\&\quad - i\omega\mu \frac{1}{r} (\partial_\theta \partial_\phi \Pi_m - \partial_\phi \partial_\theta \Pi_m).\end{aligned}$$

The Π_m contribution vanishes by the equality of mixed partial derivatives, so all magnetic terms cancel. Because the angular operator commutes with ∂_r and with multiplication by r , the surviving electric part is that same operator acting on $\frac{1}{r} \partial_r(r\pi_e)$,

$$\partial_\theta(\sin\theta E_\theta) + \partial_\phi E_\phi = \left[\partial_\theta(\sin\theta \partial_\theta) + \frac{1}{\sin\theta} \partial_\phi^2 \right] \frac{1}{r} \partial_r(r\pi_e).$$

Scattering from Dielectric Spheres (cont.)

The bracketed operator equals $\sin \theta \nabla_a^2$, where ∇_a^2 is the angular Laplacian defined explicitly below. Acting on the *full* angular mode $P_n^1(\cos \theta) \cos \phi$ (or its sine counterpart) it gives $-n(n+1) \sin \theta$ times that mode, by (4.4.28). This common angular eigenfactor is the same inside and outside the sphere. Completeness and orthogonality of the angular modes therefore force continuity of each radial coefficient, giving the first condition in (4.5.15),

$$\frac{1}{r} \partial_r(r\pi_e) \Big|_{r=a} = \frac{1}{r} \partial_r(r\pi_e^d) \Big|_{r=a}.$$

The complementary, curl-type combination of H_θ and H_ϕ from (4.4.13) cancels Π_m in the same way and isolates Π_e with the prefactor $i\omega\epsilon$,

$$\begin{aligned} \frac{1}{\sin \theta} [\partial_\theta(\sin \theta H_\phi) - \partial_\phi H_\theta] \\ = - \frac{i\omega\epsilon}{r} \nabla_a^2(r\pi_e), \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

where $\nabla_a^2 = \partial_\theta^2 + \cot \theta \partial_\theta + \frac{1}{\sin^2 \theta} \partial_\phi^2$ is the angular Laplacian. Continuity of the tangential field again removes the common eigenvalue $-n(n+1)$ of P_n^1 , leaving the algebraic condition

$$\epsilon \pi_e = \epsilon_d \pi_e^d \quad \text{at } r = a.$$

By duality $(\vec{E}, \vec{H}, \epsilon, \mu) \rightarrow (\vec{H}, -\vec{E}, \mu, \epsilon)$ the magnetic potential obeys the same pair of relations. The divergence-type combination $\partial_\theta(\sin \theta H_\theta) + \partial_\phi H_\phi$ supplies the radial-derivative continuity, while the curl-type combination of the electric field,

$$\begin{aligned} \frac{1}{\sin \theta} [\partial_\theta(\sin \theta E_\phi) - \partial_\phi E_\theta] \\ = \frac{i\omega\mu}{r} \nabla_a^2(r\pi_m), \end{aligned}$$

supplies the algebraic one, so that at $r = a$

$$\frac{1}{r} \partial_r(r\pi_m) = \frac{1}{r} \partial_r(r\pi_m^d), \quad \mu \pi_m = \mu_d \pi_m^d.$$

Scattering from Dielectric Spheres (cont.)

These four relations are the decoupled conditions (4.5.15)–(4.5.16). With this assignment, the coefficients below take the form in (4.5.23)–(4.5.24). The matching conditions ensure that each Debye potential outside the sphere couples exclusively to its counterpart inside, so each azimuthal dependence ($\cos \phi$ or $\sin \phi$) of the incident field is preserved in the scattered and internal fields. Thus, for the scattered field potential, we let

$$\pi_e^s = \frac{-E_0 \cos \phi}{k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} a_n \hat{h}_n^{(2)}(kr) P_n^1(\cos \theta), \quad (4.5.17)$$

$$\pi_m^s = \frac{-E_0 \sin \phi}{\eta k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} b_n \hat{h}_n^{(2)}(kr) P_n^1(\cos \theta), \quad (4.5.18)$$

Scattering from Dielectric Spheres (cont.)

where the Riccati–Hankel function $\hat{h}_n^{(2)}(kr) = kr h_n^{(2)}(kr)$ is used to meet the radiation condition, and the coefficients a_n and b_n are to be determined. The total potential outside the sphere is thus written as

$$\pi_e = \frac{E_0 \cos \phi}{k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} \left[\hat{j}_n(kr) - a_n \hat{h}_n^{(2)}(kr) \right] P_n^1(\cos \theta), \quad (4.5.19)$$

and

$$\pi_m = \frac{E_0 \sin \phi}{\eta k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} \left[\hat{j}_n(kr) - b_n \hat{h}_n^{(2)}(kr) \right] P_n^1(\cos \theta). \quad (4.5.20)$$

Scattering from Dielectric Spheres (cont.)

For the Debye potential inside the sphere, we let

$$\pi_e^d = \frac{E_0 \cos \phi}{k_d^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} c_n \hat{j}_n(k_d r) P_n^1(\cos \theta), \quad (4.5.21)$$

$$\pi_m^d = \frac{E_0 \sin \phi}{\eta_d k_d^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} d_n \hat{j}_n(k_d r) P_n^1(\cos \theta). \quad (4.5.22)$$

Applying the matching conditions (4.5.15) and (4.5.16) to the above equations mode by mode fixes the four unknown series. For the electric potential, the exterior and interior radial dependences per mode are (the common factor $E_0 \cos \phi (-i)^{n-1} \frac{2n+1}{n(n+1)} P_n^1(\cos \theta)$ cancels)

$$r\pi_e \sim \frac{1}{k^2} [\hat{j}_n(kr) - a_n \hat{h}_n^{(2)}(kr)], \quad r\pi_e^d \sim \frac{c_n}{k_d^2} \hat{j}_n(k_d r).$$

Scattering from Dielectric Spheres (cont.)

Continuity of $\frac{1}{r}\partial_r(r\pi_e)$ and of $\epsilon\pi_e$ from (4.5.15) then gives, at $r = a$,

$$\begin{aligned}\frac{1}{k} [\hat{j}'_n(ka) - a_n \hat{h}_n^{(2)'}(ka)] &= \frac{1}{k_d} c_n \hat{j}'_n(k_d a), \\ \frac{\epsilon}{k^2} [\hat{j}_n(ka) - a_n \hat{h}_n^{(2)}(ka)] &= \frac{\epsilon_d}{k_d^2} c_n \hat{j}_n(k_d a).\end{aligned}$$

Here and below, a prime on a Riccati–Bessel or Riccati–Hankel function means differentiation with respect to that function's complete dimensionless argument. To display the elimination explicitly, abbreviate

$$x = ka, \quad y = k_d a, \quad A = \sqrt{\epsilon\mu_d}, \quad B = \sqrt{\epsilon_d\mu},$$

and write $\hat{j}_x = \hat{j}_n(x)$, $\hat{h}_x = \hat{h}_n^{(2)}(x)$, and $\hat{j}_y = \hat{j}_n(y)$; here $x = ka$ is the Mie size parameter, distinct from the Legendre argument $x = \cos\theta$ used earlier. Since $\epsilon/k^2 = 1/(\omega^2\mu)$ and $\epsilon_d/k_d^2 = 1/(\omega^2\mu_d)$, the two electric equations above are equivalently

$$\hat{j}'_x - a_n \hat{h}'_x = c_n \frac{k}{k_d} \hat{j}'_y, \quad \hat{j}_x - a_n \hat{h}_x = c_n \frac{\mu}{\mu_d} \hat{j}_y.$$

Scattering from Dielectric Spheres (cont.)

To avoid dividing by either $\hat{j}_y$ or $\hat{j}'_y$, which can vanish at isolated arguments, cross-multiply the two equations. Using

$$\frac{\mu_d}{\mu} \frac{k}{k_d} = \sqrt{\frac{\epsilon \mu_d}{\epsilon_d \mu}} = \frac{A}{B}$$

shows that $B(k/k_d) = A(\mu/\mu_d)$. Multiplying the first equation by $B\hat{j}_y$ and the second by $A\hat{j}'_y$ therefore gives

$$B(\hat{j}'_x - a_n \hat{h}'_x) \hat{j}_y = A(\hat{j}_x - a_n \hat{h}_x) \hat{j}'_y.$$

Collecting the terms proportional to a_n therefore yields

$$a_n (A \hat{h}_x \hat{j}'_y - B \hat{h}'_x \hat{j}_y) = A \hat{j}_x \hat{j}'_y - B \hat{j}_x \hat{j}'_y,$$

Scattering from Dielectric Spheres (cont.)

which is (4.5.23). The internal coefficient can be obtained without cancelling an internal Bessel factor. Put $q = k/k_d$ and $s = \mu/\mu_d$, so $s/q = B/A$. The two electric equations are the 2×2 system

$$\begin{bmatrix} \hat{h}'_x & q\hat{j}'_y \\ \hat{h}_x & s\hat{j}_y \end{bmatrix} \begin{bmatrix} a_n \\ c_n \end{bmatrix} = \begin{bmatrix} \hat{j}'_x \\ \hat{j}_x \end{bmatrix}.$$

The Riccati–Bessel Wronskian is

$$\hat{j}_n(x)\hat{h}_n^{(2)'}(x) - \hat{j}_n'(x)\hat{h}_n^{(2)}(x) = -i.$$

If $D_e = A\hat{h}_x\hat{j}'_y - B\hat{h}'_x\hat{j}_y$ denotes the denominator of a_n , the system determinant is

$$\begin{aligned} \Delta_e &= s\hat{h}'_x\hat{j}_y - q\hat{h}_x\hat{j}'_y \\ &= -\frac{q}{A} (A\hat{h}_x\hat{j}'_y - B\hat{h}'_x\hat{j}_y) = -\frac{q}{A} D_e. \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

Cramer's rule and the Wronskian give the numerator $\hat{h}'_x \hat{j}_x - \hat{h}_x \hat{j}'_x = -i$, and hence

$$c_n = \frac{-i}{\Delta_e} = \frac{iA}{qD_e} = \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{D_e},$$

which is (4.5.25).

For the magnetic potential, retaining its different normalization $1/(\eta k^2)$ is essential. Conditions (4.5.16) reduce mode by mode to

$$\hat{j}'_x - b_n \hat{h}'_x = d_n \frac{\mu}{\mu_d} \hat{j}'_y, \quad \hat{j}_x - b_n \hat{h}_x = d_n \frac{k}{k_d} \hat{j}_y.$$

Eliminating d_n and using $(k_d/k)(\mu/\mu_d) = B/A$ gives

$$A(\hat{j}'_x - b_n \hat{h}'_x) \hat{j}_y = B(\hat{j}_x - b_n \hat{h}_x) \hat{j}'_y,$$

so

$$b_n (B \hat{h}_x \hat{j}'_y - A \hat{h}'_x \hat{j}_y) = B \hat{j}_x \hat{j}'_y - A \hat{j}'_x \hat{j}_y,$$

Scattering from Dielectric Spheres (cont.)

which is (4.5.24). With $D_m = B\hat{h}_x\hat{j}'_y - A\hat{h}'_x\hat{j}_y$, retain the same $q = k/k_d$ and $s = \mu/\mu_d$. The magnetic equations form

$$\begin{bmatrix} \hat{h}'_x & s\hat{j}'_y \\ \hat{h}_x & q\hat{j}_y \end{bmatrix} \begin{bmatrix} b_n \\ d_n \end{bmatrix} = \begin{bmatrix} \hat{j}'_x \\ \hat{j}_x \end{bmatrix}.$$

Its determinant and Cramer's numerator are

$$\Delta_m = -\frac{q}{A}D_m, \quad \hat{h}'_x\hat{j}_x - \hat{h}_x\hat{j}'_x = -i,$$

so

$$d_n = \frac{-i}{\Delta_m} = \frac{iA}{qD_m} = \frac{i\mu_d\sqrt{\epsilon_d/\mu}}{D_m},$$

which is (4.5.26). These determinant formulas remain valid when $\hat{j}_y$ or $\hat{j}'_y$ vanishes individually. They assume $D_e \neq 0$ and $D_m \neq 0$; a zero of a full determinant is a genuine modal pole and must be handled by a limiting (usually lossy or complex-frequency) formulation. As an immediate consistency check, if $\epsilon_d = \epsilon$ and $\mu_d = \mu$, then $y = x$,

Scattering from Dielectric Spheres (cont.)

both scattering numerators vanish, and the Wronskian gives $a_n = b_n = 0$ and $c_n = d_n = 1$, as required when the “sphere” is indistinguishable from its host. The resulting coefficients are

$$a_n = \frac{\sqrt{\epsilon_d \mu_d} \hat{j}_n(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d \mu_d} \hat{j}_n'(ka) \hat{j}_n(k_d a)}{\sqrt{\epsilon_d \mu_d} \hat{h}_n^{(2)}(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d \mu_d} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)}, \quad (4.5.23)$$

$$b_n = \frac{\sqrt{\epsilon_d \mu_d} \hat{j}_n(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d \mu_d} \hat{j}_n'(ka) \hat{j}_n(k_d a)}{\sqrt{\epsilon_d \mu_d} \hat{h}_n^{(2)}(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d \mu_d} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)}, \quad (4.5.24)$$

Scattering from Dielectric Spheres (cont.)

$$c_n = \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{\sqrt{\epsilon\mu_d} \hat{h}_n^{(2)}(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d \mu} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)}, \quad (4.5.25)$$

$$d_n = \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{\sqrt{\epsilon_d \mu} \hat{h}_n^{(2)}(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon\mu_d} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)}. \quad (4.5.26)$$

Let us consider the case in far field. From (4.2.17), when $kr \rightarrow \infty$, the Riccati–Hankel function reduces to a pure outgoing spherical wave. Since the asymptotic form (4.2.17) holds for arbitrary order, we apply it at $\nu = n + \frac{1}{2}$:

$$H_{n+1/2}^{(2)}(kr) \approx \sqrt{\frac{2}{\pi kr}} e^{-i\left(kr - \frac{(n+1/2)\pi}{2} - \frac{\pi}{4}\right)}.$$

Scattering from Dielectric Spheres (cont.)

Multiplying by $\sqrt{\pi kr/2}$ cancels the radical prefactor, leaving only the phase. The half-integer order shifts that phase by an extra quarter cycle,

$$-\frac{(n+1/2)\pi}{2} - \frac{\pi}{4} = -\frac{n\pi}{2} - \frac{\pi}{4} - \frac{\pi}{4} = -\frac{n\pi}{2} - \frac{\pi}{2},$$

so that

$$\begin{aligned}\hat{h}_n^{(2)}(kr) &\approx e^{-i(kr - \frac{n\pi}{2} - \frac{\pi}{2})} \\ &= e^{i\frac{n\pi}{2}} e^{i\frac{\pi}{2}} e^{-ikr} = i^n i e^{-ikr}.\end{aligned}$$

Using $e^{i\pi/2} = i$ and $e^{in\pi/2} = i^n$, the leading factor collapses to i^{n+1} , giving

$$\hat{h}_n^{(2)}(kr) = \sqrt{\frac{\pi kr}{2}} H_{n+1/2}^{(2)}(kr) \approx i^{n+1} e^{-ikr}, \quad (4.5.27)$$

Scattering from Dielectric Spheres (cont.)

we get

$$\Pi_e^s = r\pi_e^s \approx e^{-ikr} \frac{E_0 \cos \phi}{k^2} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} a_n P_n^1(\cos \theta), \quad (4.5.28)$$

$$\Pi_m^s = r\pi_m^s \approx e^{-ikr} \frac{E_0 \sin \phi}{\eta k^2} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} b_n P_n^1(\cos \theta). \quad (4.5.29)$$

Noting that in the far field

$$\partial_r \Pi_e^s \approx -ik \Pi_e^s, \quad \partial_r \Pi_m^s \approx -ik \Pi_m^s, \quad (4.5.30)$$

Scattering from Dielectric Spheres (cont.)

we can thus express E_θ and E_ϕ in the far zone explicitly. Substituting (4.4.12), replacing $\partial_r \rightarrow -ik$ through the far-field rule above, and using the angular functions π_n and τ_n defined below, the θ -component becomes

$$\begin{aligned} E_\theta &\approx \frac{-ik}{r} \partial_\theta \Pi_e^s - \frac{i\omega\mu}{r \sin \theta} \partial_\phi \Pi_m^s \\ &= \frac{-i E_0 \cos \phi}{k} \frac{e^{-ikr}}{r} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} a_n \tau_n \\ &\quad - \frac{i\omega\mu}{\eta} \frac{E_0 \cos \phi}{k^2} \frac{e^{-ikr}}{r} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} b_n \pi_n, \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

where $\partial_\phi(\sin \phi) = \cos \phi$ has been used in the magnetic term. Since $\eta = \sqrt{\mu/\epsilon}$ and $k = \omega\sqrt{\mu\epsilon}$, we have $\omega\mu/\eta = k$, so one power of k cancels and both sums carry the common factor $E_0 \cos \phi$. Collecting them,

$$E_\theta \approx \frac{-i E_0 \cos \phi}{k} \frac{e^{-ikr}}{r} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} (a_n \tau_n + b_n \pi_n),$$

which is exactly $E_0 (e^{-ikr}/r) f_\theta$ with $f_\theta = -i \cos \phi S_2/k$. An identical reduction of (4.4.12), now with $\partial_\phi(\cos \phi) = -\sin \phi$, gives

$$\begin{aligned} E_\phi &\approx \frac{-ik}{r \sin \theta} \partial_\phi \Pi_e^s + \frac{i\omega\mu}{r} \partial_\theta \Pi_m^s \\ &= \frac{i E_0 \sin \phi}{k} \frac{e^{-ikr}}{r} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} (a_n \pi_n + b_n \tau_n), \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

so that $f_\phi = i \sin \phi S_1/k$. Reading off the angular sums motivates the definitions S_1, S_2 and the functions π_n, τ_n below. In accordance with (4.1.3), E_0 multiplies the scattering amplitude and is *not* part of $\vec{f}$:

$$E_\theta \approx E_0 f_\theta(\theta, \phi) \frac{e^{-ikr}}{r}, \quad (4.5.31)$$

$$E_\phi \approx E_0 f_\phi(\theta, \phi) \frac{e^{-ikr}}{r}, \quad (4.5.32)$$

$$f_\theta(\theta, \phi) = -i \cos \phi \frac{S_2(\theta)}{k}, \quad (4.5.33)$$

$$f_\phi(\theta, \phi) = i \sin \phi \frac{S_1(\theta)}{k}, \quad (4.5.34)$$

$$S_1(\theta) = \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} [a_n \pi_n(\cos \theta) + b_n \tau_n(\cos \theta)], \quad (4.5.35)$$

Scattering from Dielectric Spheres (cont.)

$$S_2(\theta) = \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} [a_n \tau_n(\cos \theta) + b_n \pi_n(\cos \theta)], \quad (4.5.36)$$

with

$$\pi_n(\cos \theta) = \frac{P_n^1(\cos \theta)}{\sin \theta}, \quad \tau_n(\cos \theta) = \frac{d}{d\theta} P_n^1(\cos \theta). \quad (4.5.37)$$

By (4.1.5), the differential cross section is

$$\sigma_d(\theta, \phi) = |\vec{f}(\theta, \phi)|^2 = \left| \cos \phi \frac{S_2(\theta)}{k} \right|^2 + \left| \sin \phi \frac{S_1(\theta)}{k} \right|^2, \quad (4.5.38)$$

and by (4.1.8), the scattering cross section is

$$\sigma_{sca} = \int_{4\pi} \sigma_d(\theta, \phi) d\Omega = \frac{\pi}{k^2} \int_0^\pi (|S_2(\theta)|^2 + |S_1(\theta)|^2) \sin \theta d\theta. \quad (4.5.39)$$

Scattering from Dielectric Spheres (cont.)

Computing the following:

$$\begin{aligned} |S_1(\theta)|^2 &= \sum_{n=1}^{\infty} \sum_{n'=1}^{\infty} \frac{2n+1}{n(n+1)} \frac{2n'+1}{n'(n'+1)} \\ &\quad \times [a_n \pi_n a_{n'}^* \pi_{n'} + b_n \tau_n b_{n'}^* \tau_{n'} \\ &\quad + a_n \pi_n b_{n'}^* \tau_{n'} + b_n \tau_n a_{n'}^* \pi_{n'}], \end{aligned} \quad (4.5.40)$$

$$\begin{aligned} |S_2(\theta)|^2 &= \sum_{n=1}^{\infty} \sum_{n'=1}^{\infty} \frac{2n+1}{n(n+1)} \frac{2n'+1}{n'(n'+1)} \\ &\quad \times [a_n \tau_n a_{n'}^* \tau_{n'} + b_n \pi_n b_{n'}^* \pi_{n'} \\ &\quad + a_n \tau_n b_{n'}^* \pi_{n'} + b_n \pi_n a_{n'}^* \tau_{n'}], \end{aligned} \quad (4.5.41)$$

Scattering from Dielectric Spheres (cont.)

Adding the two expansions and grouping by amplitude pair, the angular factors combine into $\pi_n \pi_{n'} + \tau_n \tau_{n'}$ for the like-amplitude products ($a_n a_{n'}^*$ and $b_n b_{n'}^*$) and into $\pi_n \tau_{n'} + \tau_n \pi_{n'}$ for the mixed products ($a_n b_{n'}^*$ and $b_n a_{n'}^*$):

$$\begin{aligned} \int_0^\pi (|S_1|^2 + |S_2|^2) \sin \theta \, d\theta &= \sum_{n=1}^{\infty} \sum_{n'=1}^{\infty} \frac{2n+1}{n(n+1)} \frac{2n'+1}{n'(n'+1)} \\ &\times \left[(a_n a_{n'}^* + b_n b_{n'}^*) \int_0^\pi (\pi_n \pi_{n'} + \tau_n \tau_{n'}) \sin \theta \, d\theta \right. \\ &\quad \left. + (a_n b_{n'}^* + b_n a_{n'}^*) \int_0^\pi (\pi_n \tau_{n'} + \tau_n \pi_{n'}) \sin \theta \, d\theta \right]. \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

These angular integrals satisfy the orthogonality relations

$$\int_0^\pi (\pi_n \pi_{n'} + \tau_n \tau_{n'}) \sin \theta d\theta = \begin{cases} 0, & n \neq n', \\ \frac{2n^2(n+1)^2}{2n+1}, & n = n', \end{cases} \quad (4.5.42)$$

$$\int_0^\pi (\pi_n \tau_{n'} + \tau_n \pi_{n'}) \sin \theta d\theta = 0.$$

The second relation eliminates every mixed term, while the first collapses the like-amplitude terms to the diagonal $n' = n$. For that diagonal, the coefficient simplifies as

$$\left(\frac{2n+1}{n(n+1)} \right)^2 \frac{2n^2(n+1)^2}{2n+1} = 2(2n+1),$$

Scattering from Dielectric Spheres (cont.)

and with $a_n a_n^* = |a_n|^2$ and $b_n b_n^* = |b_n|^2$ we obtain

$$\begin{aligned}\sigma_{\text{sca}} &= \frac{\pi}{k^2} \int_0^\pi (|S_1|^2 + |S_2|^2) \sin \theta \, d\theta \\ &= \frac{2\pi}{k^2} \sum_{n=1}^{\infty} (2n+1) (|a_n|^2 + |b_n|^2) .\end{aligned}\tag{4.5.43}$$

For a Rayleigh sphere, *both* dimensionless size parameters must be small:

$$|x| = ka \ll 1, \quad |y| = |k_d|a \ll 1.$$

The equality $|x| = ka$ uses the lossless-host choice $k > 0$; the modulus on y is essential when the passive interior wavenumber k_d is complex. Define the dimensionless material wavenumber ratio at the start of the derivation by

$$\kappa \equiv \frac{y}{x} = \frac{k_d}{k}, \quad y = \kappa x.$$

Scattering from Dielectric Spheres (cont.)

Thus κ is not the spherical-harmonic azimuthal order m . For fixed material parameters, assume that all electric and magnetic quasistatic multipole denominators are nonzero:

$$n\epsilon_d + (n+1)\epsilon \neq 0, \quad n\mu_d + (n+1)\mu \neq 0, \quad n = 1, 2, \dots$$

Then the multipoles with $n > 1$ are of higher order, and the $n = 1$ electric and magnetic terms give the leading scattering. If a material contrast is instead allowed to vary with x , while the remaining normalized material parameters stay bounded and κ remains finite and nonzero, its normalized quasistatic detuning must remain asymptotically large compared with x^2 :

$$\left| \frac{\epsilon_d + 2\epsilon}{\epsilon} \right| \gg x^2, \quad \left| \frac{\mu_d + 2\mu}{\mu} \right| \gg x^2.$$

At detuning $O(x^2)$, the $O(x^2)$ dynamic correction to the denominator is of the same order as its quasistatic part and must be retained. These detuning conditions control the dipole approximations only; they do not exclude higher-multipole resonances when the

Scattering from Dielectric Spheres (cont.)

contrasts vary with x . For fixed material contrast, with A and B as defined above, substituting the small-argument expansions

$$\hat{j}_1(z) = \frac{z^2}{3} + O(z^4), \quad \hat{j}_1'(z) = \frac{2z}{3} + O(z^3),$$
$$\hat{h}_1^{(2)}(x) = \frac{i}{x} + O(x), \quad \hat{h}_1^{(2)'}(x) = -\frac{i}{x^2} + O(1)$$

into the $n = 1$ numerator and denominator of (4.5.23) gives every leading term explicitly:

$$\begin{aligned} N_a &= A \hat{j}_1(x) \hat{j}_1'(y) - B \hat{j}_1'(x) \hat{j}_1(y) \\ &= A \left(\frac{x^2}{3} \right) \left(\frac{2\kappa x}{3} \right) - B \left(\frac{2x}{3} \right) \left(\frac{\kappa^2 x^2}{3} \right) + O(x^5) \\ &= \frac{2\kappa x^3}{9} (A - B\kappa) + O(x^5), \\ D_a &= A \hat{h}_1^{(2)}(x) \hat{j}_1'(y) - B \hat{h}_1^{(2)'}(x) \hat{j}_1(y) \\ &= A \left(\frac{i}{x} \right) \left(\frac{2\kappa x}{3} \right) - B \left(-\frac{i}{x^2} \right) \left(\frac{\kappa^2 x^2}{3} \right) + O(x^2) \\ &= \frac{i\kappa}{3} (2A + B\kappa) + O(x^2). \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

Because the quasistatic denominator is nonzero by hypothesis, division is valid. Cancelling the common nonzero factor κ , and using $1/i = -i$, gives

$$\begin{aligned} a_1 = \frac{N_a}{D_a} &= \frac{2\kappa x^3 (A - B\kappa)/9}{i\kappa(2A + B\kappa)/3} + O(x^5) \\ &= \frac{2x^3}{3i} \frac{A - B\kappa}{2A + B\kappa} + O(x^5) \\ &= \frac{2i}{3} x^3 \frac{B\kappa - A}{2A + B\kappa} + O(x^5). \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

The magnetic coefficient follows from (4.5.24) with the same steps:

$$\begin{aligned}N_b &= B\hat{j}_1(x)\hat{j}'_1(y) - A\hat{j}'_1(x)\hat{j}_1(y) \\&= \frac{2\kappa x^3}{9}(B - A\kappa) + O(x^5), \\D_b &= B\hat{h}_1^{(2)}(x)\hat{j}'_1(y) - A\hat{h}_1^{(2)'}(x)\hat{j}_1(y) \\&= \frac{i\kappa}{3}(2B + A\kappa) + O(x^2), \\b_1 &= \frac{N_b}{D_b} = \frac{2x^3}{3i} \frac{B - A\kappa}{2B + A\kappa} + O(x^5) \\&= \frac{2i}{3}x^3 \frac{A\kappa - B}{2B + A\kappa} + O(x^5).\end{aligned}$$

Scattering from Dielectric Spheres (cont.)

It remains to remove the square roots without silently changing branches. The correlated roots specified after (4.5.1) obey

$$\kappa^2 = \frac{\epsilon_d \mu_d}{\epsilon \mu}, \quad \frac{A}{B} = \frac{\mu_d}{\mu \kappa}.$$

Consequently,

$$\frac{B\kappa}{A} = \frac{\kappa^2 \mu}{\mu_d} = \frac{\epsilon_d}{\epsilon}, \quad \frac{A\kappa}{B} = \frac{\mu_d}{\mu}.$$

Substitution in the two ratios therefore gives

$$\begin{aligned} a_1 &= \frac{2i}{3}(ka)^3 \frac{\epsilon_d - \epsilon}{\epsilon_d + 2\epsilon} + O((ka)^5), \\ b_1 &= \frac{2i}{3}(ka)^3 \frac{\mu_d - \mu}{\mu_d + 2\mu} + O((ka)^5). \end{aligned} \tag{4.5.44}$$

When the corresponding contrast numerator is nonzero, these expressions show that the leading dimensionless Mie coefficient scales as $(ka)^3$. If $\epsilon_d = \epsilon$ or $\mu_d = \mu$, the associated

Scattering from Dielectric Spheres (cont.)

cubic term vanishes and, if that coefficient is needed, its next nonzero order must be retained. If both equalities hold, the sphere is identical to its host and all scattering coefficients vanish exactly, as established above. Keeping the $n = 1$ terms in $\sigma_{\text{sca}} = \frac{2\pi}{k^2} \sum_n (2n+1)(|a_n|^2 + |b_n|^2)$ and substituting (4.5.44) gives the complete leading result under the fixed-contrast nonresonance assumptions above

$$\begin{aligned}\sigma_{\text{sca}} &= \frac{2\pi}{k^2} 3 (|a_1|^2 + |b_1|^2) + O(k^6 a^8) \\ &= \frac{8\pi}{3} k^4 a^6 \left[\left| \frac{\epsilon_d - \epsilon}{\epsilon_d + 2\epsilon} \right|^2 + \left| \frac{\mu_d - \mu}{\mu_d + 2\mu} \right|^2 \right] + O(k^6 a^8).\end{aligned}$$

The material ratios in brackets are dimensionless, so the prefactor $k^4 a^6$ has the required cross-section dimension of area. Thus the often-quoted ω^4 (or λ^{-4}) law additionally assumes that the host and particle material parameters are approximately frequency independent over the band of interest. For ordinary nonmagnetic molecular matter, this leading electric-dipole behavior explains why air molecules scatter blue light more

Scattering from Dielectric Spheres (cont.)

strongly than red light, why direct sunlight becomes redder along the long atmospheric path near sunset, and why the lunar sky remains black in the absence of an atmosphere. The scattering from a PEC sphere can be obtained as a limiting case of the general dielectric Mie solution. Specifically, with μ_d held finite, real, and positive, let the permittivity $\epsilon_d \rightarrow -i\infty$. The matching conditions then enforce that the tangential electric field vanishes at the sphere surface, and the Mie coefficients simplify considerably. To see how this limit acts on (4.5.23)–(4.5.26), note that as $\epsilon_d \rightarrow -i\infty$ the interior wavenumber $k_d = \omega\sqrt{\epsilon_d\mu_d}$ also diverges. In each coefficient the two competing terms scale as $\sqrt{\epsilon_d\mu_d}$ and $\sqrt{\epsilon_d\mu}$. Since $|\sqrt{\epsilon_d\mu}| \rightarrow \infty$ while $\sqrt{\epsilon_d\mu_d}$ stays finite, only the $\sqrt{\epsilon_d\mu}$ terms survive in the numerator and denominator of a_n and b_n . For a_n these are the primed factors $\hat{j}'_n(ka)$ and $\hat{h}_n^{(2)'}(ka)$, sharing the common factor $\sqrt{\epsilon_d\mu}\hat{j}_n(k_d a)$,

$$a_n \rightarrow \frac{-\sqrt{\epsilon_d\mu}\hat{j}'_n(ka)\hat{j}_n(k_d a)}{-\sqrt{\epsilon_d\mu}\hat{h}_n^{(2)'}(ka)\hat{j}_n(k_d a)} = \frac{\hat{j}'_n(ka)}{\hat{h}_n^{(2)'}(ka)}.$$

Scattering from Dielectric Spheres (cont.)

For b_n the surviving terms instead carry the unprimed factors $\hat{j}_n(ka)$, $\hat{h}_n^{(2)}(ka)$ together with the common factor $\sqrt{\epsilon_d \mu} \hat{j}'_n(k_d a)$,

$$b_n \rightarrow \frac{\sqrt{\epsilon_d \mu} \hat{j}_n(ka) \hat{j}'_n(k_d a)}{\sqrt{\epsilon_d \mu} \hat{h}_n^{(2)}(ka) \hat{j}'_n(k_d a)} = \frac{\hat{j}_n(ka)}{\hat{h}_n^{(2)}(ka)},$$

so the diverging factors cancel and the finite results below remain. For the internal coefficients the numerator grows at most like $\sqrt{\epsilon_d}$, while the denominator carries $\hat{j}_n(k_d a)$ or $\hat{j}'_n(k_d a)$ with $k_d a \rightarrow \infty$. The branch $\epsilon_d \rightarrow -i\infty$ fixes $k_d a \rightarrow |k_d a| e^{-i\pi/4}$, so that, with the $e^{+i\omega t}$ convention adopted here, $\hat{j}_n(k_d a)$ grows without bound; the denominator therefore dominates and

$$c_n \sim \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{-\sqrt{\epsilon_d \mu} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)} \rightarrow 0,$$

$$d_n \sim \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{\sqrt{\epsilon_d \mu} \hat{h}_n^{(2)}(ka) \hat{j}'_n(k_d a)} \rightarrow 0.$$

Scattering from Dielectric Spheres (cont.)

Collecting these limits gives

$$a_n = \frac{\hat{j}'_n(ka)}{\hat{h}_n^{(2)'}(ka)}, \quad (4.5.45)$$

$$b_n = \frac{\hat{j}_n(ka)}{\hat{h}_n^{(2)}(ka)}, \quad (4.5.46)$$

$$c_n = 0, \quad (4.5.47)$$

$$d_n = 0. \quad (4.5.48)$$

All coefficients of the internal fields vanish, so the PEC solution is a special case embedded within the more general dielectric formulation. To obtain the monostatic curve without skipping its angular reduction, first use the phase-free convention $P_n^1(\cos \theta) = \sin \theta P'_n(\cos \theta)$, where here $P'_n(u) = dP_n(u)/du$. Using u as the

Scattering from Dielectric Spheres (cont.)

Legendre-polynomial argument keeps it distinct from the Mie size parameter $x = ka$. The Legendre equation at $u = 1$, together with $P_n(1) = 1$, gives

$$-2P'_n(1) + n(n+1)P_n(1) = 0, \quad P'_n(1) = \frac{n(n+1)}{2}.$$

Parity, $P_n(-u) = (-1)^n P_n(u)$, then gives $P'_n(-1) = (-1)^{n+1} n(n+1)/2$. Consequently, at backscatter $\theta = \pi$,

$$\begin{aligned}\pi_n(-1) &= P'_n(-1) = (-1)^{n+1} \frac{n(n+1)}{2}, \\ \tau_n(-1) &= -P'_n(-1) = (-1)^n \frac{n(n+1)}{2}.\end{aligned}$$

Substitution into the angular sums gives

$$\begin{aligned}S_1(\pi) &= \frac{1}{2} \sum_{n=1}^{\infty} (2n+1)(-1)^n (b_n - a_n), \\ S_2(\pi) &= -S_1(\pi).\end{aligned}$$

Scattering from Dielectric Spheres (cont.)

Although the individual spherical unit vectors depend on the arbitrary azimuth at the pole, the squared amplitude in (4.1.5) does not:

$$\sigma_d(\pi) = \frac{|S_1(\pi)|^2}{k^2}.$$

Using $\sigma_{\text{mono}} = 4\pi\sigma_d(\pi)$ and writing the electrical size as $x = ka = 2\pi a/\lambda$ therefore yields the dimensionless result used for the plot,

$$\frac{\sigma_{\text{mono}}}{\pi a^2} = \frac{1}{x^2} \left| \sum_{n=1}^{\infty} (2n+1)(-1)^n [a_n - b_n] \right|^2, \quad x = ka. \quad (4.5.49)$$

Figure 4.1 plots (4.5.49) against $x = ka$, not against a/λ ; the conversion is $a/\lambda = x/(2\pi)$. In the large-sphere physical-optics limit, $\sigma_{\text{mono}} \rightarrow \pi a^2$, so this normalization tends to unity.

Scattering from Dielectric Spheres

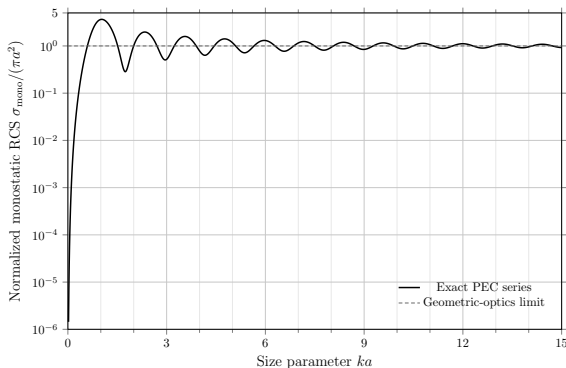


Figure 4.1: Normalized monostatic RCS $\sigma_{\text{mono}}/(\pi a^2)$ of a PEC sphere versus the size parameter $x = ka = 2\pi a/\lambda$.

Further Reading

For a concise and accessible overview of special functions, the reader is encouraged to consult Novak and Fox [20]. A more comprehensive mathematical treatment can be found in Arfken, Weber, and Harris' *Mathematical Methods for Physicists* [1], which provides detailed derivations and properties of Bessel, Legendre, and other special functions widely used in physics and engineering. The *NIST Digital Library of Mathematical Functions (DLMF)* [9] serves as an authoritative online reference for the definitions, identities, and asymptotic expansions of special functions. Harrington's text [11] includes dedicated chapters on cylindrical and spherical wave solutions. In this work, we adopt the same selection of canonical examples—conducting cylinders and dielectric spheres—as in Zangwill's treatment [31], since these configurations serve as representative models for physical objects such as wires and raindrops. Ishimaru's classic [12] provides a rigorous formulation of Mie theory, which forms the principal reference for this chapter. For a broader discussion of electromagnetic scattering phenomena, the reader is referred to Osipov and Tretyakov's *Modern Electromagnetic Scattering Theory with Applications* [22]. In the context of light scattering by small particles, classical treatments include Van de Hulst's *Light Scattering by Small Particles* [29], Bohren and Huffman's *Absorption and Scattering of Light by Small Particles* [4], and Kerker's *Scattering of Light and Other Electromagnetic Radiation* [16].

Problems

- 1 Using (4.3.1) and (4.3.6), plot the well-defined normalized magnitude $|E_z^t(\rho, \phi)|/E_0$ versus $k\rho \geq 1$ for $ka = 1$ along the fixed rays $\phi = 0, \pi/2$, and π . State the series truncation used and verify numerically that $E_z^t = 0$ at $k\rho = ka = 1$.
- 2 Verify directly that (4.3.10)–(4.3.12) satisfy Faraday's law and the source-free Ampère–Maxwell law term by term.
- 3 For $ka = 1$, evaluate the TM surface current from both (4.3.13) and (4.3.14); demonstrate numerical agreement for a stated series truncation.
- 4 For the TE solution (4.3.22), evaluate (4.3.25) at $ka = 1$ and verify numerically that $E_\phi^t(a, \phi) = 0$.
- 5 Independently reproduce the mode-by-mode elimination leading to (4.5.23)–(4.5.26), including the Riccati–Bessel Wronskian used for c_n and d_n .
- 6 Verify (4.5.44).

References



George B. Arfken, Hans J. Weber, and Frank E. Harris.
Mathematical Methods for Physicists: A Comprehensive Guide.
Academic Press, 7th edition, 2011.
Electronic edition.



Constantine A. Balanis.
Advanced Engineering Electromagnetics.
Wiley, 2nd edition, 2012.



Kasra Barkeshli.
Advanced Electromagnetics and Scattering Theory.
Springer, 1st edition, 2015.
Edited by Sina Khorasani.



Craig F. Bohren and Donald R. Huffman.
Absorption and Scattering of Light by Small Particles.
Wiley Science Series. John Wiley & Sons, 2008.
Revised electronic reprint of the 1983 original.

References (cont.)



Weng Cho Chew.

Waves and Fields in Inhomogeneous Media.

IEEE Press Series on Electromagnetic Wave Theory. Wiley-IEEE Press, 1st edition, 1999.

IEEE Press reprint of the 1990 version with corrections and minor updating.



Phillip C. Clemmow.

The Plane Wave Spectrum Representation of Electromagnetic Fields, volume 12 of *International Series of Monographs in Electromagnetic Waves*.

Elsevier Science, 2013.

Electronic reissue of the 1966 Pergamon Press first edition.



Robert E. Collin.

Field Theory of Guided Waves.

IEEE Press, 2nd edition, 1990.

References (cont.)



Georges A. Deschamps.

Electromagnetics and differential forms.

Proceedings of the IEEE, 69(6):676–696, June 1981.



NIST Digital Library of Mathematical Functions.

<https://dlmf.nist.gov/>, Release 1.2.7 of 2026-06-15.

F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.



Stuart Gregson, John McCormick, and Clive Parini.

Principles of Planar Near-Field Antenna Measurements.

Number 53 in IET Electromagnetic Waves Series. Institution of Engineering and Technology, 2007.



Roger F. Harrington.

Time-Harmonic Electromagnetic Fields.

IEEE Press Series on Electromagnetic Wave Theory. IEEE Press, 2001.

Classic reissue of the 1961 original.

References (cont.)



Akira Ishimaru.

Electromagnetic Wave Propagation, Radiation, and Scattering: From Fundamentals to Applications.

Wiley-IEEE Press, 2nd edition, 2017.



John David Jackson.

Classical Electrodynamics.

Wiley India, 3rd edition, 2012.

Indian reprint of the third edition.



Shyh-Kang Jeng.

Scattering from a cavity-backed slit in a ground plane—TE case.

IEEE Transactions on Antennas and Propagation, 38(10):1523–1529, October 1990.



Jian-Ming Jin.

Theory and Computation of Electromagnetic Fields.

Wiley-IEEE Press, 2nd edition, 2015.

References (cont.)



Milton Kerker.

The Scattering of Light and Other Electromagnetic Radiation, volume 16 of *Physical Chemistry: A Series of Monographs*.

Academic Press, 2013.

Series edited by Ernest M. Loebl; electronic reissue of the 1969 first edition.



Jin Au Kong.

Theory of Electromagnetic Waves.

Wiley-Interscience, New York, 1975.



Jin Au Kong.

Electromagnetic Wave Theory.

EMW Publishing, Cambridge, MA, 2008.



Ismo V. Lindell.

Differential Forms in Electromagnetics.

Wiley-IEEE Press, 2004.

References (cont.)



Kyle A. Novak and Laura J. Fox.

Special Functions of Mathematical Physics: A Tourist's Guidebook.

Equal Share Press, LLC, 2019.



Lukas Novotny and Bert Hecht.

Principles of Nano-Optics.

Cambridge University Press, 2nd edition, 2012.



Andrey V. Osipov and Sergei A. Tretyakov.

Modern Electromagnetic Scattering Theory with Applications.

John Wiley & Sons, 1st edition, 2017.



Prabhakar H. Pathak and Robert J. Burkholder.

Electromagnetic Radiation, Scattering, and Diffraction.

Wiley-IEEE Press, 2021.

References (cont.)



Xin-Qing Sheng.

Electromagnetic Theory, Calculations, and Applications.

Tsinghua University Press, Beijing, 2nd edition, 2023.

In Chinese.



Xin-Qing Sheng and Wei Song.

Essentials of Computational Electromagnetics.

Wiley, 2012.



Glenn S. Smith.

An Introduction to Classical Electromagnetic Radiation.

Cambridge University Press, 1st edition, 1997.



Carlo G. Someda.

Electromagnetic Waves.

CRC Press, 2nd edition, 2006.

References (cont.)



Chen-To Tai.

General Vector and Dyadic Analysis: Applied Mathematics in Field Theory.
Wiley-IEEE Press, 2nd edition, 1997.



H. C. van de Hulst.

Light Scattering by Small Particles.
Dover Publications, 1981.
Reprint of the 1957 Wiley edition.



Pochi Yeh.

Optical Waves in Layered Media.
Wiley Series in Pure and Applied Optics. Wiley, 2005.
Paperback reissue of the 1988 original.



Andrew Zangwill.

Modern Electrodynamics.
Cambridge University Press, 2013.